Math Inquirers

Extension Workbook

with answers

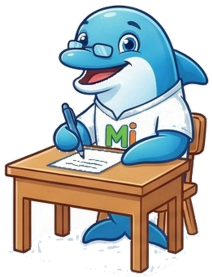
Level 1

Self-study | Ages 6-8



My name _____

Welcome!



This book is for you to explore at your own pace. Every unit has 7 pages. You do not need a teacher.

For the child

1. Read **Learn it**. Look at the worked example.
2. Do **Start here**. Write in the boxes and on the lines.
3. Try **Think harder**. Stuck? Read the hint. It is upside down!
4. Try **Go further** when you want a big challenge.
5. Do **Check yourself** and count your score.

For the grown-up

- Practice for the same ideas as the Holiday Program, with new numbers and stories.
- About 3 short sessions a week, 20 to 25 minutes each (see the pace plan).
- Let your child try before looking at the answers.
- Answers are at the back, from page 77, with a short reason for each. Read the reason together.
- Check yourself score 2 or less? Repeat Learn it and one Start here page.

The four labels

● **Start here. The main path. Do these first.**

◆ **Think harder. Longer problems. Hints are upside down.**

★ **Go further. Big challenges, just for fun.**

✓ **Check yourself. Four quick questions at the end.**

Tips

- Use a pencil. Mistakes help you learn.
- Read each question twice before you write.
- Think harder and Go further are not a race. Come back another day.

Contents

1	**Number patterns** Learn it, practice and check · answers on page 78	page 5
2	**Number bonds** Learn it, practice and check · answers on page 79	page 12
3	**Picture graphs** Learn it, practice and check · answers on page 81	page 19
4	**Repeating and growing patterns** Learn it, practice and check · answers on page 82	page 26
5	**Working backwards** Learn it, practice and check · answers on page 84	page 33
6	**Money** Learn it, practice and check · answers on page 86	page 40
7	**2D shapes** Learn it, practice and check · answers on page 88	page 47
8	**Logic and sorting** Learn it, practice and check · answers on page 89	page 54
9	**Length** Learn it, practice and check · answers on page 91	page 61
10	**Puzzle day** Learn it, practice and check · answers on page 93	page 68
	My progress	page 75
	My certificate	page 76
	Answers For grown-ups, with short reasons	page 77

My pace plan

One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. Tick the box when the week is done.

Week	Unit	Session 1	Session 2	Session 3	Done
1	**Number patterns**	Pages 5-6 Learn it, Start here	Pages 7-8 Start here	Pages 9-11 Think harder, Go further, Check yourself	☐
2	**Number bonds**	Pages 12-13 Learn it, Start here	Pages 14-15 Start here	Pages 16-18 Think harder, Go further, Check yourself	☐
3	**Picture graphs**	Pages 19-20 Learn it, Start here	Pages 21-22 Start here	Pages 23-25 Think harder, Go further, Check yourself	☐
4	**Repeating and growing patterns**	Pages 26-27 Learn it, Start here	Pages 28-29 Start here	Pages 30-32 Think harder, Go further, Check yourself	☐
5	**Working backwards**	Pages 33-34 Learn it, Start here	Pages 35-36 Start here	Pages 37-39 Think harder, Go further, Check yourself	☐
6	**Money**	Pages 40-41 Learn it, Start here	Pages 42-43 Start here	Pages 44-46 Think harder, Go further, Check yourself	☐
7	**2D shapes**	Pages 47-48 Learn it, Start here	Pages 49-50 Start here	Pages 51-53 Think harder, Go further, Check yourself	☐
8	**Logic and sorting**	Pages 54-55 Learn it, Start here	Pages 56-57 Start here	Pages 58-60 Think harder, Go further, Check yourself	☐
9	**Length**	Pages 61-62 Learn it, Start here	Pages 63-64 Start here	Pages 65-67 Think harder, Go further, Check yourself	☐
10	**Puzzle day**	Pages 68-69 Learn it, Start here	Pages 70-71 Start here	Pages 72-74 Think harder, Go further, Check yourself	☐
11	**Catch-up week**	Repeat any Check yourself where your score was 2 or less.			☐

Tips for the grown-up

- Short and often works better than one long session. Stop while your child still wants more.

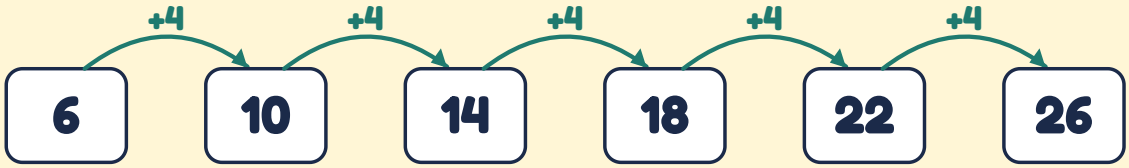
Learn it: Number patterns

My goal. I can find the rule of a number pattern and use it to write the next numbers.

A number pattern follows a rule. The rule tells you how to get from one number to the next. A frog hopping along a number strip makes a pattern.

Worked example

Kumar's frog starts on 6. It hops 4 each time. Write the first 6 numbers.



1. Start on 6. Write 6.
2. Hop 4. 6 and 4 more is 10.
3. Hop 4 again. 10 and 4 more is 14.
4. Keep going: 18, 22, 26. Each number is 4 more than the number before.

Answer: 6, 10, 14, 18, 22, 26. The rule is add 4.

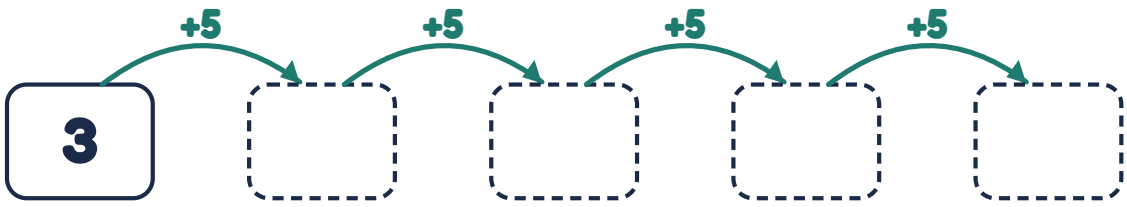
How to find a rule

1. Look at the first 2 numbers. How far is it from one to the other?
2. Check the next 2 numbers. Is the hop the same?
3. Going up? The rule is add. Going down? The rule is take away.
4. Use the rule to write the next numbers.

Your turn

Answer at the back of the book.

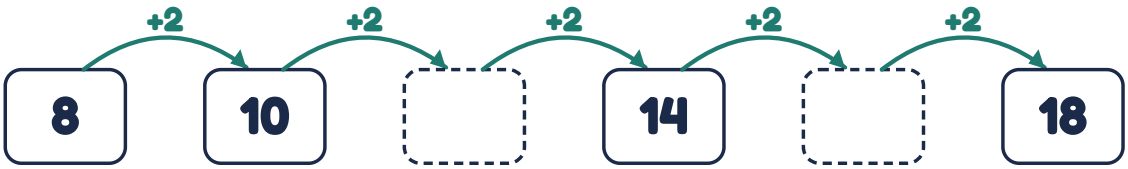
Try **A frog starts on 3 and hops 5 each time. Write the missing numbers in the tiles.**



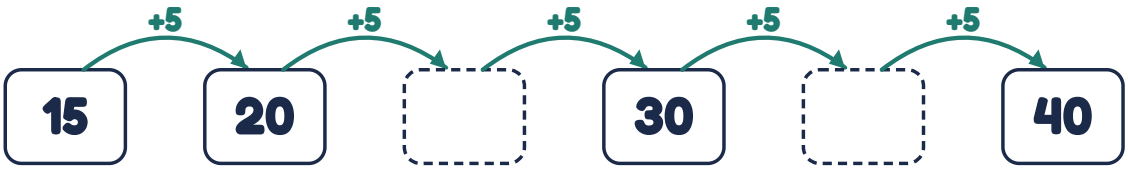
Start here

Do these first. Take your time.

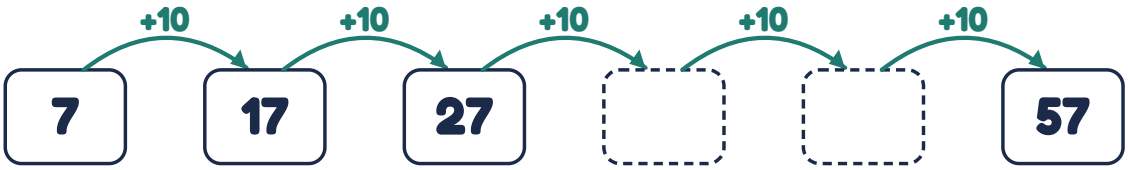
1 Count in 2s. Write the missing numbers in the tiles.



2 Count in 5s. Write the missing numbers in the tiles.



3 Count in 10s. Write the missing numbers in the tiles.



4 Fill in the blanks.

(a) 5, 9, 13, 17, ☐, ☐

(b) The rule is add ☐.

5 Fill in the blanks.

(a) 21, 18, 15, 12, ☐, ☐

(b) The rule is take away ☐.

Start here

Keep going. Check your work.

6 **Circle the rule for this pattern: 7, 11, 15, 19.**

add 3 add 4 add 5

7 **Look at the pattern: 30, 26, 22, 18, ...**

(a) Do the numbers go up or down? Up Down

(b) The rule is take away ☐.

(c) The next number is ☐.

8 **Siti puts 3 stickers in box 1, 6 stickers in box 2 and 9 stickers in box 3. She keeps going.**

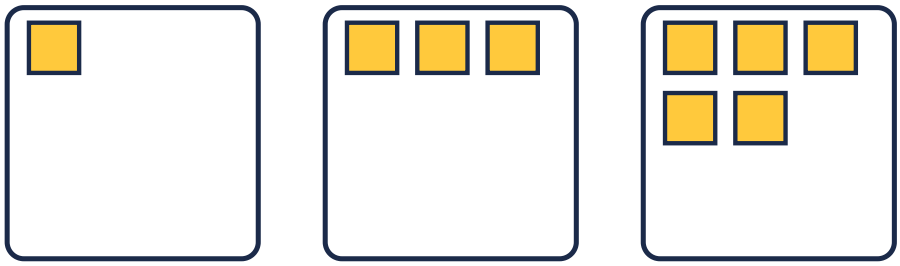
(a) How many stickers go in box 5? ☐ stickers

(b) How many stickers go in box 6? ☐ stickers

Write the numbers: _____

Hint. Each box has 3 more than the one before.

9 **Ben builds steps with blocks.**



Step 1 Step 2 Step 3

(a) How many blocks are in Step 4? ☐ blocks

(b) Each step has ☐ more blocks than the step before.

Start here

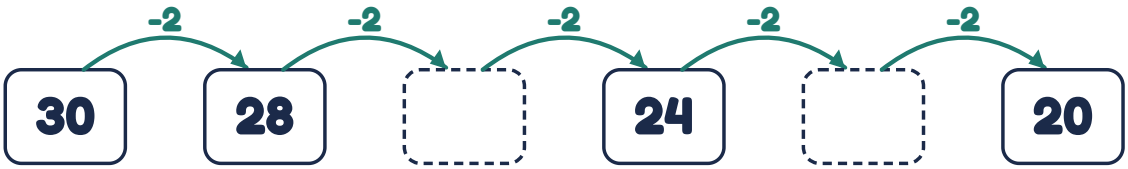
Nearly there. You can do it.

10 One number is wrong. Circle it. Then write the right number.

9, 12, 15, 17, 21, 24

The right number is ☐.

11 Count back in 2s. Write the missing numbers in the tiles.



12 Wei puts 5¢ in his jar on Monday, 10¢ on Tuesday and 15¢ on Wednesday.

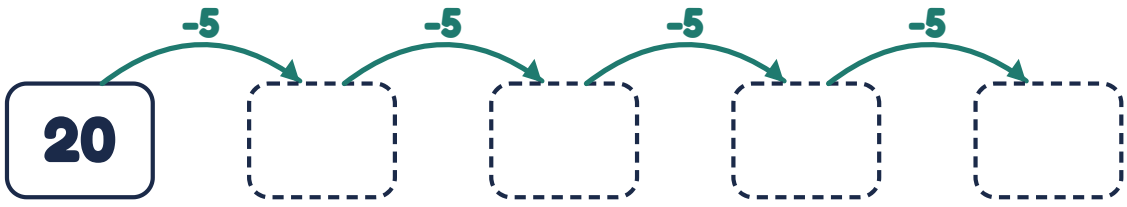
He keeps going. How much does he put in on Friday? ☐ ¢

Write the days and amounts: _____

13 Mei starts on 3 and hops 4 each time. Circle every number she lands on.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31

14 Kumar starts on 20 and counts back in 5s. Write the missing numbers in the tiles.



Think harder

Hints are upside down. Try first!

1 **Siti starts on 2 and hops 5 each time.**

(a) Will she land on 37? Yes No

(b) Will she land on 40? Yes No

Write the numbers she lands on: _____

_____

Hint. Write the numbers she lands on, one hop at a time. Stop when you pass 40.

2 **Ali counts in 4s from 4 up to 50. Mei counts in 10s from 10 up to 50.**

(a) Which numbers do they both say? _____

(b) How far apart are the numbers they both say? ☐

Ali: _____

Mei: _____

Hint. Write Ali's numbers on one line and Mei's numbers on the other. Look for matches.

3 **A pattern adds the same number each time. Its 2nd number is 10. Its 5th number is 25.**

(a) How many hops from the 2nd number to the 5th number? ☐

(b) The rule is add ☐.

(c) The 1st number is ☐.

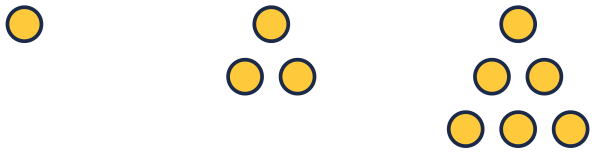
Hint. From 10 to 25 is 15. Share 15 equally between the hops.

★ Go further

Big challenges. Hints are upside down.

★ Beyond Level 1: Triangle numbers

1 Look at the triangles made of dots. Triangle 1 has 1 dot, Triangle 2 has 3 dots and Triangle 3 has 6 dots. The pattern keeps going.



Triangle 1 Triangle 2 Triangle 3

(a) Triangle 4 has ☐ dots.

(b) Triangle 5 has ☐ dots.

(c) Triangle 7 has ☐ dots.

Hint. How many more dots does each triangle have than the one before? Is it always the same?

2 Mei starts on 1 and adds 3 each time. Ali starts on 2 and adds 5 each time. Both stop at 40. Write the numbers that Mei and Ali both say.

Mei: _____

Ali: _____

Both: _____

Hint. Write each pattern on its own line. Then look for the same number in both lines.

3 A pattern starts on 2 and adds the same whole number each time. The pattern says 20. Which numbers can it add? Find them all.

I tried: _____

_____

Hint. 20 is 18 more than 2. Which equal hops make exactly 18?

✔ Check yourself

No hints. Try each one on your own.

1 14, 18, 22, 26, ☐, ☐

Working: _____

2 40, 35, 30, 25, ☐, ☐. The rule is take away ☐.

Working: _____

3 Is 30 in this pattern: 6, 10, 14, 18, ...? Yes No

Working: _____

4 One number is wrong: 11, 16, 21, 27, 31, 36. The right number is ☐.

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a grown-up. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

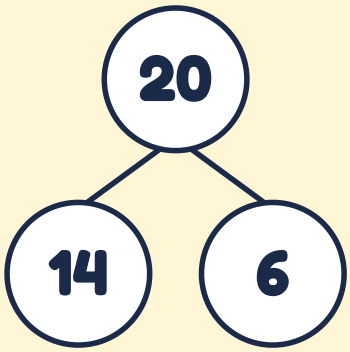
Learn it: Number bonds

My goal. I can find the missing partner in a bond to 10 and to 20. I can tell when the 2 sides are equal.

A number bond is 2 numbers that make a whole. 6 and 4 make 10. So 6 + 4 = 10 and 10 = 6 + 4. Bonds to 10 help us find bonds to 20.

Worked example

Aisha has 20 beads. 14 beads are blue and the rest are white. How many beads are white?



20 = 14 + 6

1. The whole is 20. One part is 14. We want the other part.
2. 14 is 10 and 4 more. We need 4 and some more to make 10: 4 + 6 = 10.
3. So 14 + 6 = 20. The other part is 6.
4. Check: 14 + 6 = 20.

Answer: 6 beads are white. 20 = 14 + 6.

How to find the missing partner

1. Find the whole (the greatest number).
2. Look at the ones digit of the part you know. What makes 10 with it?
3. Add the partners. Do they make the whole?

Your turn

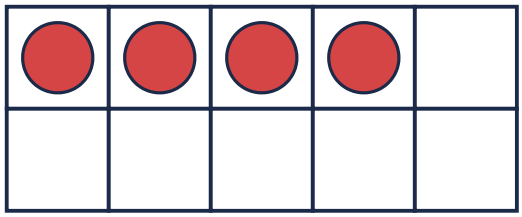
Answer at the back of the book.

Try **Fill in the blank. 20 = 17 + ☐**

Start here

Do these first. Take your time.

1 **Look at the ten-frame.**



(a) How many more counters fill the frame? ☐ counters

(b) 10 = 4 + ☐

2 **Fill in the missing numbers.**

(a) 10 = 8 + ☐

(b) 10 = ☐ + 3

(c) 10 = 5 + ☐

(d) 10 = 1 + ☐

3 **Kumar has 10 marbles. 6 marbles are in his pocket. The rest are in a bag.**

How many marbles are in the bag? ☐ **marbles**

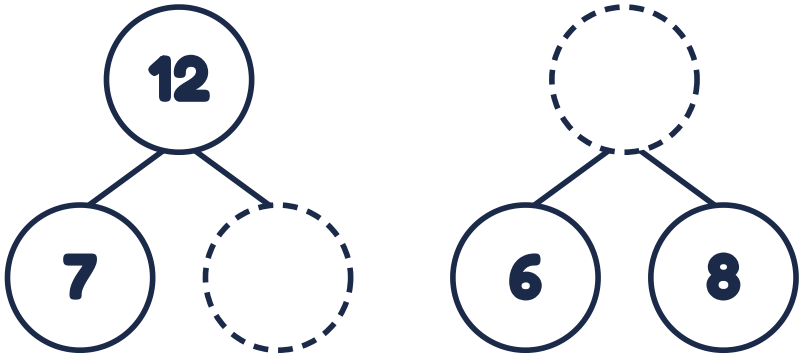
4 **Write the partner that makes 10.**

1 and ☐ **3 and** ☐ **6 and** ☐ **8 and** ☐

Start here

Keep going. Check your work.

5 Write the missing number in each bond.



6 Fill in the missing numbers.

(a) 20 = 15 + ☐

(b) 20 = ☐ + 14

(c) 20 = 18 + ☐

(d) 20 = 11 + ☐

Hint. Look at the ones digit. What makes 10 with it?

7 7 + 3 = 10. Use it to fill in the blanks.

17 + ☐ = 20 27 + ☐ = 30 47 + ☐ = 50

8 Mei has a shelf for 20 books. There are 16 books on the shelf. How many more books fit? ☐ books

Start here

Nearly there. You can do it.

9 **Fill in the missing number. Both sides must be equal.**

(a) 6 + 5 = 7 + ☐

(b) 8 + 2 = 5 + ☐

(c) 9 + 4 = ☐ + 5

10 **Look at the pairs that make 9.**

(a) 1 + 8 = 9 2 + ☐ = 9 3 + ☐ = 9 4 + ☐ = 9

(b) The first numbers go up by 1. The second numbers go up down by 1.

11 **Circle True or False.**

(a) 4 + 6 = 10 True False

(b) 8 + 3 = 10 True False

(c) 13 + 7 = 20 True False

(d) 15 + 6 = 20 True False

12 **There are 20 children in a club. 7 children are girls. The rest are boys. How many boys are in the club?** ☐ **boys**

13 **Kumar has 8 stickers.**

(a) He wants 10 stickers. How many more does he need? ☐ stickers

(b) He wants 20 stickers. How many more does he need? ☐ stickers

Think harder

Hints are upside down. Try first!

1 **Ravi puts 14 marbles into 2 bags. The first bag has fewer marbles than the second bag. Each bag has at least 1 marble. Write every way in the table.**

How many ways are there? ☐

First bag	Second bag

Hint. Try 1 marble in the first bag, then 2. Stop when the first bag is not fewer.

2 **Each shape stands for a number. The same shape is always the same number.**

(a) ★ + ★ + 5 = 17. ★ = ☐

(b) ● + ● + ● = 21. ● = ☐

Hint. Take 5 away from 17 first. Then share what is left between the 2 stars.

3 **Three different numbers from 1 to 6 add up to 9. Find every set of three.**

☐ + ☐ + ☐ = 9

☐ + ☐ + ☐ = 9

☐ + ☐ + ☐ = 9

Hint. Write the numbers from smallest to greatest. Start with 1 as the smallest number.

★ Go further

Big challenges. Hints are upside down.

★ Beyond Level 1: Making two amounts equal

1 **Ali has 16 stickers. Mei has 8 stickers. Ali gives some of his stickers to Mei. Now each child has an equal number of stickers.**

(a) How many stickers does Ali give Mei? ☐ stickers

(b) How many stickers does each child have now? ☐ stickers

Hint. Put all the stickers together. Share them equally between the 2 children.

2 **Mei puts 17 sweets in 2 jars. Jar A has 5 more sweets than Jar B.**

(a) Jar A has ☐ sweets.

(b) Jar B has ☐ sweets.

Hint. Take away the 5 extra sweets. Share what is left between the 2 jars.

3 **I am a number. When I add 7, the answer ends in 0 and is between 15 and 45. What can I be? Find them all.**

The answer can be: _____

So I can be: _____

Hint. First write the numbers between 15 and 45 that end in 0.

4 **Pick 2 different numbers from 1, 2, 3, 4, 5 and add them.**

(a) Write every total you can make. Begin with the smallest.

_____

(b) Which totals can you make in only one way? _____

My pairs: _____

_____

Hint. List the pairs in order: 1 + 2, 1 + 3, 1 + 4, 1 + 5, then 2 + 3 …

✔ Check yourself

No hints. Try each one on your own.

1 $10 = 2 + \square$

Working: _____

2 $20 = 6 + \square$

Working: _____

3 $7 + 5 = 9 + \square$

Working: _____

4 There are 20 flowers. 11 are red. The rest are yellow. How many flowers are yellow? ☐ flowers

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a grown-up. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

Learn it: Picture graphs

My goal. I can read a picture graph and use it to add and compare.

A picture graph uses pictures to show how many. The key tells us what 1 picture means. Count the pictures in a row.

Worked example

The graph shows books borrowed in 1 week. How many more books on Tuesday than on Thursday?

Books borrowed

Monday	📖 📖 📖 📖
Tuesday	📖 📖 📖 📖 📖 📖 📖
Wednesday	📖 📖 📖 📖 📖
Thursday	📖 📖

Key: 📖 **= 1 book**

1. Read the key. 1 picture is 1 book.
2. Count the Tuesday row: 7 books. Count the Thursday row: 2 books.
3. "How many more" asks for the gap. Subtract: 7 - 2 = 5. Check: 2 + 5 = 7.

Answer: 5 more books are borrowed on Tuesday. 7 - 2 = 5.

How to read a picture graph

1. Read the title and the key.
2. Count the pictures in each row you need.
3. Add for "in all". Subtract for "how many more".

Your turn

Answer at the back of the book.

Try **Look at the graph. How many more shells does Ben have than Kumar?**

☐ shells

Ben	★ ★ ★ ★ ★
Kumar	★ ★

Key: ★ **= 1 shell**

Start here

Do these first. Take your time.

1 **Look at the graph. Fill in the blanks.**

Games at recess

Skipping	☺☺☺☺☺☺☺
Tag	☺☺☺☺☺☺☺
Catch	☺☺☺☺☺
Hopscotch	☺☺☺☺
Marbles	☺☺☺

Key: ☺ = 1 child

(a) ☐ children chose catch.

(b) Which game did the fewest children choose? _____

(c) Which 2 games got the same number of votes?

_____

2 **Look at the graph in Question 1.**

(a) How many more children chose skipping than hopscotch? ☐ children

(b) How many children are there altogether? ☐ children

3 **Look at the graph in Question 1. Circle True or False.**

(a) More children chose catch than hopscotch. True False

(b) Marbles got 4 votes. True False

4 **Siti asks 5 friends which game they like best. 4 friends choose tag. Is it true that most children in the school like tag best?**

Yes **No** **We cannot tell**

Start here

Keep going. Check your work.

5 Look at the graph. Fill in the blanks.

Stickers this week

Ali	★★★★★★
Mei	★★★★★★★★★
Ben	★★★
Siti	★★★★★

Key: ★ = 1 sticker

(a) Who has the most stickers? _____

(b) Ben has ☐ stickers.

(c) Mei has ☐ more stickers than Siti.

6 Look at the graph in Question 5.

(a) How many stickers do Ali and Ben have in all? ☐ stickers

(b) Ali gives 2 stickers to Ben. How many stickers does Ben have now? ☐ stickers

7 Draw 1 face for each child. 7 children chose water, 4 chose milk and 2 chose tea. How many faces did you draw in all? ☐ faces

Drinks at the party

Water	☐☐☐☐☐☐☐☐
Milk	☐☐☐☐☐☐☐☐
Tea	☐☐☐☐☐☐☐☐

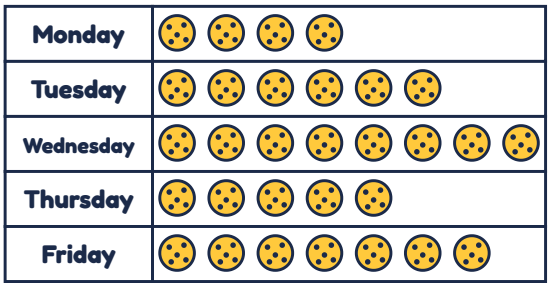
Key: ☺ = 1 child

Start here

Nearly there. You can do it.

8 Look at the graph. Fill in the blanks.

Cookies sold at the fair



Key: = 1 cookie

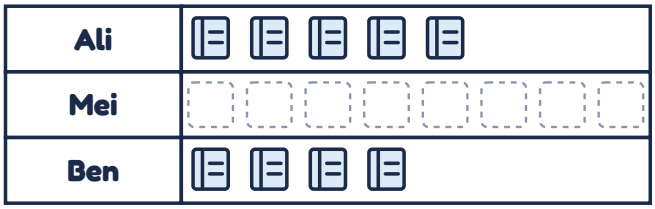
(a) On which day were the most cookies sold? _____

(b) ☐ cookies were sold on Thursday.

(c) How many more cookies were sold on Friday than on Monday?

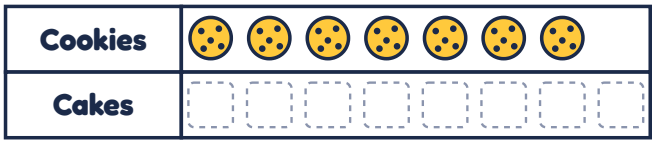
☐ cookies

9 Ali has 5 books. Mei has 3 more books than Ali. Draw Mei's row. How many books does Mei have? ☐ books



Key: = 1 book

10 Ravi makes 12 treats in all. 7 are cookies. The rest are cakes. Draw the cakes. How many cakes does Ravi make? ☐ cakes



Key: = 1 treat

Think harder

Hints are upside down. Try first!

★ Beyond Level 1: Keys bigger than 1

1 Each face stands for 2 children. A half face stands for 1 child.

Children in clubs

Drawing	☺ ☺ ☺ ☺
Singing	☺ ☺ ☺
Chess	☺ ☺ ☺ ☺ ☺ ◔

Key: ☺ = 2 children ◔ = 1 child

(a) How many children chose chess? ☐ children

(b) How many more children chose chess than singing? ☐ children

Hint. Count in 2s along each row. A half face is 1 more child.

2 30 children vote for a class trip. Zoo gets 9 votes, museum 6 and beach 8. The rest vote for the farm.

(a) How many children chose the farm? ☐ children

(b) Which place got the most votes? _____

Hint. Add the 3 numbers you know. Then count on to 30.

★ Beyond Level 1: Comparing two keys

3 Look at the 2 graphs. Their keys are not the same.

Graph P	☺ ☺ ☺ ☺ ☺ ☺

Key: ☺ = 1 child

Graph Q	☺ ☺

Key: ☺ = 5 children

(a) Graph P shows ☐ children. Graph Q shows ☐ children.

(b) Which graph shows more children? Graph P Graph Q

Hint. Do not count the pictures only. Read the key first.

★ Go further

Big challenges. Hints are upside down.

★ Beyond Level 1: Build a graph from clues

1 **Ravi, Lily, Jamie and Ray find 24 shells in all. Ravi finds 7. Lily finds 2 fewer than Ravi. Jamie and Ray find the same number.**

Ravi	★★★★★★★
Lily	
Jamie	
Ray	

Key: ★ = 1 shell

(a) Lily finds ☐ shells.

(b) Jamie finds ☐ shells.

Hint. Find Lily first. Then share what is left.

2 **A graph has rows for kites, balls and ropes. There are 18 toys. There are 2 more balls than kites. There is 1 more rope than kites.**

(a) Kites: ☐

(b) Balls: ☐

(c) Ropes: ☐

Hint. Take away the extra balls and the extra rope. Share what is left between 3 rows.

3 **In Wei's graph, each picture stands for 2 stickers. Wei has 6 pictures. Siti has the same number of stickers as Wei. In Siti's graph, each picture stands for 3 stickers.**

(a) How many stickers does Wei have? ☐ stickers

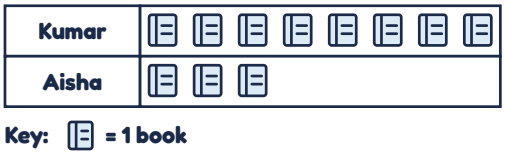
(b) How many pictures does Siti draw? ☐ pictures

Hint. Count in 2s to find the stickers. Then count in 3s.

✔ Check yourself

No hints. Try each one on your own.

1 Use the graph for Questions 1 to 4. Kumar read ☐ books.



2 How many more books did Kumar read than Aisha? ☐ books

Working: _____

3 How many books did the 2 children read in all? ☐ books

Working: _____

4 Who read the fewest books? _____

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a grown-up. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

Learn it: Repeating and growing patterns

My goal. I can find the core of a pattern and use it to find a shape far along. I can follow a growing pattern and find the rule.

A repeating pattern has a core that goes round and round. We can count in the size of the core to find a far place. A growing pattern gets larger by the same amount each time.

Worked example

The pattern keeps going. What is the 14th shape?



1. Find the core. It is triangle, circle, circle: 3 shapes.
2. Count in 3s: 3, 6, 9, 12. Place 12 is the end of a core.
3. Places 13 and 14 start a new core. Place 13 is a triangle and place 14 is a circle.

Answer: The 14th shape is a circle.

How to find a far place

1. Find the core and count its shapes.
2. Count in that size. Stop at or before your place.
3. Count on from the last full core.

Your turn

Answer at the back of the book.

Try **The pattern keeps going. What is the 10th shape?**

circle square triangle star



Name _____

Start here

Do these first. Take your time.

1 Circle the shape that comes next. triangle circle



2 Draw the next 3 shapes in the boxes. Then circle the 9th shape.

star heart



3 Dark beads and white beads make a pattern. What is the 12th bead?

dark white



4 Write the next 2 letters. M, N, O, M, N, O, M, _____

_____

5 Ravi claps and stomps. The pattern is clap, clap, stomp, clap, clap, stomp, and so on. What is the 10th action? clap stomp

6 The pattern keeps going. What is the 16th shape?

square star circle



Start here

Keep going. Check your work.

7 **Look at the pattern. Circle the right shape.**



(a) The 9th shape is a triangle circle square .

(b) The 12th shape is a triangle circle square .

(c) The 13th shape is a triangle circle square .

8 **Look at the pattern in Question 7. The circles are in places 2, 5 and 8.**

Write the next 3 places that have a circle. ☐ ☐ ☐

9 **A pattern repeats every 4 shapes. Circle True or False.**

(a) The 8th shape is the same as the 4th shape. True False

(b) The 10th shape is the same as the 4th shape. True False

(c) The 13th shape is the same as the 1st shape. True False

10 **The pattern keeps going. How many circles are in the first 12 shapes?**

☐ **circles**



11 **Lily puts 3 beads on a string on day 1. She adds 2 beads each day.**

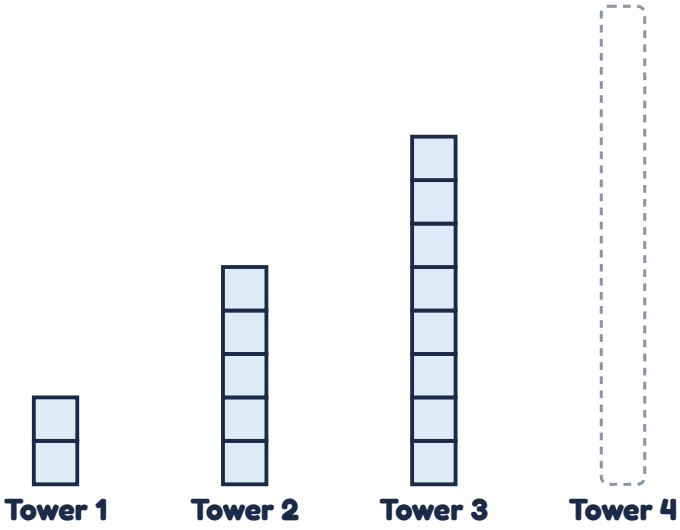
(a) How many beads does she have on day 5? ☐ beads

(b) On which day does she have 15 beads? Day ☐

Start here

Nearly there. You can do it.

12 **The towers grow in a pattern. Draw Tower 4. Fill in the blanks.**

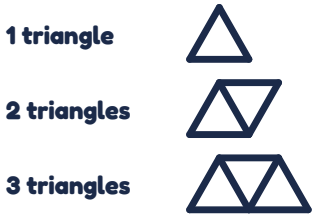


(a) Tower 4 has ☐ blocks.

(b) Tower 6 has ☐ blocks.

(c) The tower with 23 blocks is Tower ☐.

13 **Aisha makes a row of triangles with straws.**



(a) Count the straws. 2 triangles use ☐ straws.

(b) 3 triangles use ☐ straws.

(c) 4 triangles use ☐ straws.

Name _____

Think harder

Hints are upside down. Try first!

1 **The pattern keeps going.**



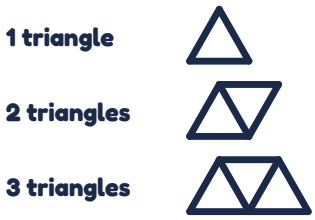
(a) What is the 15th shape? heart star

(b) How many stars are in the first 18 shapes? ☐ stars

(c) In which place is the 5th star? Place ☐

Hint. Count in 3s for the cores. Each core has 2 stars. Write the places of the stars: 2, 3, 5 ...

2 **Siti has 24 straws. She makes a row of triangles.**



(a) How many triangles can she make? ☐ triangles

(b) How many straws are left? ☐ straw

Hint. The 1st triangle uses 3 straws. Each new triangle uses 2 more. Keep counting: 3, 5, 7 ...

3 **Mei has 5 stickers on day 1, 8 stickers on day 2 and 11 stickers on day 3. The pattern keeps going.**

(a) How many stickers does she have on day 6? ☐ stickers

(b) On which day does she have 32 stickers? Day ☐

Hint. Make a table. Write the day and the stickers. Add 3 each time.

★ Go further

Big challenges. Hints are upside down.

★ Beyond Level 1: Jumps that grow

1 **Look at the number pattern: 1, 2, 4, 7, 11. The jumps are not the same.**

(a) Write the next 2 numbers. ☐ ☐

(b) What is the 8th number in the pattern? ☐

Hint. Write the jumps: +1, +2, +3 … What is the next jump?

★ Beyond Level 1: Two ways to build

2 **Ravi makes 3 triangles that do not touch. Mei makes a row of 3 triangles that share sides.**

Ravi △ △ △

Mei 

(a) Ravi needs ☐ straws for 7 triangles.

(b) Mei needs ☐ straws for 7 triangles.

(c) Mei uses ☐ fewer straws than Ravi for 7 triangles.

Hint. Ravi uses 3 straws for each triangle. Mei starts with 3 and adds 2 for each new triangle.

★ Beyond Level 1: Two patterns together

3 **Mei lays 2 patterns, one above the other. The top pattern is circle, star, circle, star. The bottom pattern is A, B, C, A, B, C.**



(a) The first place with a star above a C is place ☐.

(b) The next place with a star above a C is place ☐.

Hint. Stars are in places 2, 4, 6 … Cs are in places 3, 6, 9 … Which place is in both lists?

✔ Check yourself

No hints. Try each one on your own.

1 The pattern is sun, moon, moon, sun, moon, moon. The 9th one is

sun moon

Working: _____

2 The letters P, Q, R, S go round and round. The 18th letter is

P Q R S

Working: _____

3 Tower 1 has 3 blocks. Each tower adds 4. Tower 5 has ☐ blocks.

Working: _____

4 1 triangle uses 3 straws. Each new one adds 2. 6 triangles use ☐ straws.

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a grown-up. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

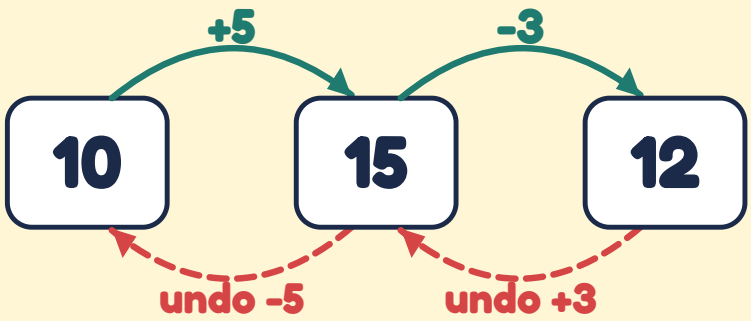
Learn it: Working backwards

My goal. I can find the start of a story when I know the end. I go backwards and undo each step.

In a mystery number story, we know the end. We undo each step.

Worked example

Ravi has some stickers. He gets 5 more, then gives 3 to Mei. Now he has 12 stickers. How many stickers did Ravi begin with?



1. Go backwards. Undo -3 with +3: 12 + 3 = 15.
2. Undo +5 with -5: 15 - 5 = 10.
3. Check going forwards: 10 + 5 = 15 and 15 - 3 = 12.

Answer: Ravi had 10 stickers at first.

How to work backwards

1. Find the end number. This is the number you know.
2. Undo the last step first. Add becomes take away. Take away becomes add.
3. Keep going back until you reach the start.
4. Check by going forwards.

Your turn

Answer at the back of the book.

Try **Mei buys 6 beads. Now she has 15 beads. How many beads did she begin with? ☐ beads**

Start here

Do these first. Take your time.

1 A cup hides some counters. 5 more counters go in. Now there are 13 counters. How many counters were in the cup at first? ☐ counters

2 Mei picks a secret number. Fill in her secret number each time.

(a) She adds 8 and gets 13. Her number is ☐.

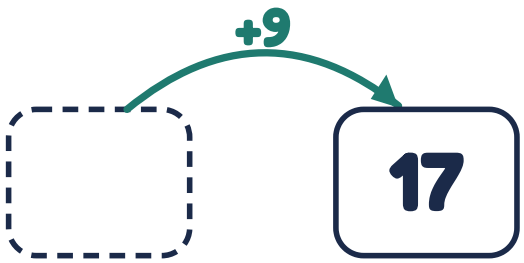
(b) She adds 7 and gets 15. Her number is ☐.

(c) She takes away 5 and gets 9. Her number is ☐.

(d) She takes away 6 and gets 12. Her number is ☐.

3 Aisha has some biscuits. She eats 4 biscuits. 9 biscuits are left. How many biscuits did she have at first? ☐ biscuits

4 Write the number in the empty box. Go backwards along the arrow.

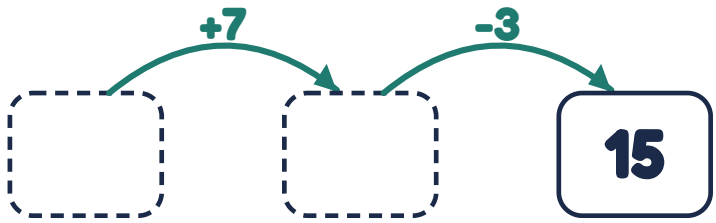


5 Siti has some pencils. She gives 3 to Ali, then buys 8 more. Now she has 14 pencils. How many pencils did she begin with? ☐ pencils

Start here

Keep going. Check your work.

6 Write the 2 missing numbers in the empty boxes. Go backwards.



7 This machine always adds the same number. Look at the table.

IN	6	9		15
OUT	10		17	

(a) The machine adds ☐.

(b) Write the 3 missing numbers in the table.

8 Fill in the missing number. Go backwards.

(a) ☐ + 6 = 15

(b) ☐ − 4 = 9

(c) 18 − ☐ = 11

9 A lift goes down 4 floors. Then it goes up 6 floors. It stops on floor 11. On which floor did the lift begin? Floor ☐

Start here

Nearly there. You can do it.

10 Kumar is 6th in a queue. There are 2 children behind him. How many children are there in the queue altogether? ☐ children

Draw the queue: _____

11 Aisha stands 5th from the front in a line of 12 children. How many children are behind her? ☐ children

Draw the queue: _____

12 Think about more and fewer.

(a) Ali has 15 stickers. He has 6 more than Siti. Siti has ☐ stickers.

(b) Ben has 9 sweets. He has 6 fewer than Jenny. Jenny has ☐ sweets.

13 Check going forwards. Circle True or False.

(a) Start at 9. Add 4. Take away 6. You get 7. True False

(b) Start at 12. Take away 5. Add 3. You get 9. True False

(c) Start at 8. Add 7. Take away 4. You get 11. True False

14 Kumar picks a number and adds 6. He gets 15. Is his number 21?

(a) Circle Yes or No. Yes No

(b) His number is ☐.

Think harder

Hints are upside down. Try first!

1 **Mei has some stickers. She gives 5 to Ali, gets 8 from Ben and gives 4 to Siti. Now she has 13 stickers.**

(a) Before she gave 4 to Siti, she had ☐ stickers.

(b) Before Ben gave her 8, she had ☐ stickers.

(c) At first Mei had ☐ stickers.

Hint. Go back one step at a time. Start with the last step: giving 4 to Siti.

2 **A machine adds 9 and then takes away 4. A one-step machine does the same job.**

(a) The one-step machine adds ☐.

(b) IN 11 gives OUT ☐.

(c) OUT 30 comes from IN ☐.

Hint. Try IN 10. Add 9, then take away 4. What do you get? How far is that from 10?

3 **Ravi is 7th from the front of a queue. He is also 4th from the back.**

(a) ☐ children are in front of Ravi.

(b) ☐ children are behind Ravi.

(c) The queue has ☐ children.

Hint. Do not forget Ravi himself. Draw the queue as a row of circles.

★ Go further

Big challenges. Hints are upside down.

★ Beyond Level 1: Undoing a double

1 **Aisha picks a number and adds it to itself. This is a double. Find each number.**

(a) 6 + 6 = ☐

(b) Double it, then add 4. She gets 18. Her number is ☐.

(c) Double it, then take away 5. She gets 11. Her number is ☐.

Hint. Undo the last step first. Then ask: which number added to itself makes this?

2 **Does the order of the steps matter?**

(a) Start at 12. Add 6, then take away 9. You get ☐.

(b) Start at 12. Take away 9, then add 6. You get ☐.

(c) Start at 5. Add 6, then take away 9. You get ☐.

(d) Start at 5. Can you take away 9 first? Yes No

Hint. Do each one a step at a time. In part (d), can you take 9 away from 5?

3 **A frog is at the bottom of a well with 12 steps. Each day it climbs 4 steps. Each night it slides back 2 steps. On which day does the frog get out?**

(a) The frog gets out on day ☐.

Day 1: _____

Day 2: _____

Day 3: _____

Hint. Write the step after each day. Then write it after each night.

✔ Check yourself

No hints. Try each one on your own.

1 Wei picks a number and adds 7. He gets 16. His number is ☐.

Working: _____

2 Lily gives away 5 stickers, then gets 9. Now she has 17. At first she had ☐ stickers.

Working: _____

3 A machine takes away 6. OUT is 9. IN is ☐.

Working: _____

4 Ali picks a number and takes away 7. He gets 8. His number is ☐.

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a grown-up. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

Learn it: Money

My goal. I can count coins and notes and pay an exact amount. I can find all the ways and the fewest coins.

Our coins are 5¢, 10¢, 20¢ and 50¢. A $1 coin is worth 100¢. To pay an amount, start with the largest coin that fits.

Worked example

Ben wants to pay exactly 75¢. He has 5¢, 10¢, 20¢ and 50¢ coins. Which coins use the fewest coins?



1. Take the largest coin that fits: 50¢. 75¢ - 50¢ = 25¢ is left.
2. The largest coin that fits 25¢ is 20¢. 25¢ - 20¢ = 5¢ is left.
3. Take a 5¢ coin. Nothing is left.
4. Check: 50¢ + 20¢ + 5¢ = 75¢. That is 3 coins.

Answer: 50¢, 20¢ and 5¢. That is 3 coins.

How to pay an exact amount

1. Take the largest coin that is not too much.
2. Subtract it from the amount. What is left?
3. Repeat with what is left. Stop at 0.
4. Add your coins to check.

Your turn

Answer at the back of the book.

Try **Pay exactly 80¢ with 3 coins. Use 5¢, 10¢, 20¢ or 50¢ coins. Write the coins.**

☐ ¢ + ☐ ¢ + ☐ ¢

Start here

Do these first. Take your time.

1 Count the money. Write the total. ☐ ¢



2 Count the money. Write the total. $ ☐



3 Fill in how many coins.

(a) 50¢ can be changed for ☐ coins of 10¢.

(b) 20¢ can be changed for ☐ coins of 5¢.

(c) $1 can be changed for ☐ coins of 20¢.

(d) $1 can be changed for ☐ coins of 50¢.

4 Pay exactly with 2 coins. Use 5¢, 10¢, 20¢ or 50¢ coins. Write the 2 coins.

(a) 25¢ = ☐ ¢ + ☐ ¢

(b) 60¢ = ☐ ¢ + ☐ ¢

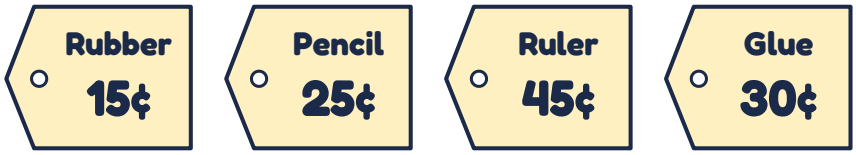
(c) 30¢ = ☐ ¢ + ☐ ¢

5 Two of me make 40¢. Which coin am I? 5¢ 10¢ 20¢ 50¢

Start here

Keep going. Check your work.

6 Look at the tags.



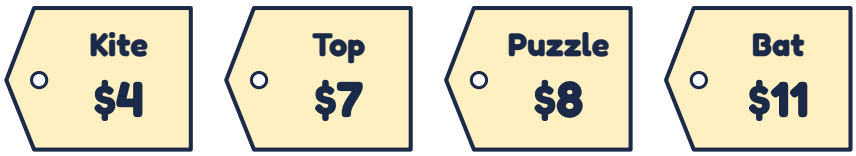
(a) A rubber and a pencil cost ☐ ¢ altogether.

(b) Which costs more, the ruler or the glue? ruler glue

(c) The ruler costs ☐ ¢ more than the rubber.

7 Ben pays 50¢ for a 30¢ sticker. How much change does he get? ☐ ¢

8 Ali buys 2 toys for exactly $12. Which 2 toys? Circle them.



9 Lily has $8. A book costs $13. How much more does she need? $ ☐

10 Look at the 2 purses.



(a) Which purse has more money? A B

(b) It has ☐ ¢ more.

Start here

Nearly there. You can do it.

11 **Make 40¢ with only 20¢ and 10¢ coins. Write every way in the table. Each row is one way.**

(a) Write the ways in the table.

(b) There are ☐ ways.

20¢ coins	10¢ coins

12 **Circle True or False.**

(a) Two 20¢ coins and one 10¢ coin make 50¢. True False

(b) Three 10¢ coins make 40¢. True False

(c) A $1 coin is worth ten 10¢ coins. True False

(d) Four 5¢ coins make 25¢. True False

13 **Pay with the fewest coins. Use 5¢, 10¢, 20¢ or 50¢ coins.**

(a) 40¢ needs ☐ coins.

(b) 90¢ needs ☐ coins.

14 **Circle the coins that pay exactly 65¢.**



Think harder

Hints are upside down. Try first!

1 **Make 65¢ with exactly 4 coins. Use 5¢, 10¢, 20¢ or 50¢ coins. Find every set.**

Set 1: _____

Set 2: _____

Hint. Try the 50¢ coin first. What is left to make with 3 coins?

2 **Siti has 7 coins. They are 5¢ coins and 10¢ coins. Together they are worth 50¢.**

(a) She has ☐ coins of 5¢.

(b) She has ☐ coins of 10¢.

5¢ coins	10¢ coins	Total

Hint. Guess how many 5¢ coins. The other coins are 10¢. Find the total. Change your guess.

3 **Mei buys exactly 2 toys for $15. There are 2 ways.**



(a) The Car and the Top Ball Kite Drum

(b) The Top and the Car Ball Kite Drum

Hint. Pick one toy. How much money is left for the second toy?

★ Go further

Big challenges. Hints are upside down.

★ Beyond Level 1: Sharing money equally

1 Friends share money equally.

(a) 2 friends share 80¢. Each gets ☐ ¢.

(b) 4 friends share 80¢. Each gets ☐ ¢.

(c) Siti and Wei share 50¢. They have only 20¢ and 10¢ coins. Can each child get the same amount? Yes No

Half of 50¢: _____

Hint. What is half of 50¢? Can 20¢ and 10¢ coins make that amount?

2 Wei pays with a $50 note for a bag and a hat. The hat costs $18. He gets $12 change.

(a) Wei spends $ ☐ altogether.

(b) The bag costs $ ☐.

Spent: _____

Bag: _____

Hint. Change is the money that is left. How much did Wei really spend?

3 Ali pays exactly $38. He has $1 coins and $2, $5 and $10 notes. He uses the fewest notes and coins.

(a) He uses ☐ notes and coins.

(b) They are: _____

Try: _____

Hint. Use as many $10 notes as you can. Then look at what is left.

✔ Check yourself

No hints. Try each one on your own.

1 Count the money. Write the total. ☐ ¢



2 Pay exactly 70¢ with 3 coins. ☐ ¢ + ☐ ¢ + ☐ ¢

Working: _____

3 Ravi has 90¢. A drink costs 65¢. How much is left? ☐ ¢

Working: _____

4 Which is more money? 85¢ $1

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a grown-up. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

Learn it: 2D shapes

My goal. I can name shapes by their sides and corners. I can count shapes, squares and triangles in a picture.

Shapes have straight sides and corners. A triangle has 3 sides and 3 corners. A circle has none. A square and a rectangle have 4 of each.

Worked example

Look at the shapes. How many shapes have 4 corners?



1. A shape with 4 corners has 4 straight sides.
2. Check each shape. A square has 4 corners. A rectangle has 4 corners.
3. There are 2 squares and 1 rectangle.
4. The other shapes do not have 4 corners.

Answer: 3 shapes.

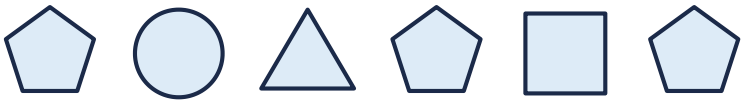
How to count shapes

1. Pick one kind of shape.
2. Touch each one once. Make a small mark on it.
3. Write down how many.
4. Do the next kind. Add all the counts to check.

Your turn

Answer at the back of the book.

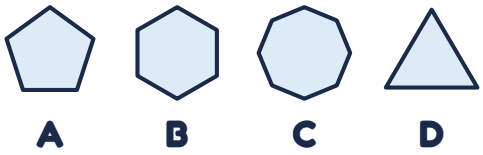
Try **How many pentagons are in the picture? A pentagon has 5 sides.**



Start here

Do these first. Take your time.

1 **Look at the 4 shapes. Circle the right name for each shape.**



(a) Shape A is a pentagon hexagon octagon

(b) Shape B is a pentagon hexagon octagon

(c) Shape C is an pentagon hexagon octagon

(d) Shape D is a triangle square circle

2 **Fill in the blanks.**

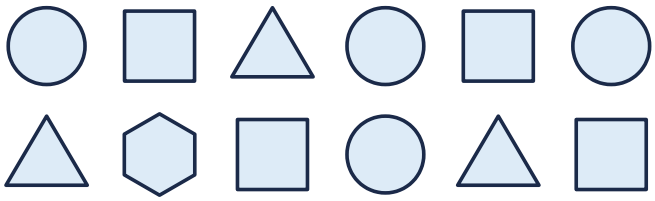
(a) A pentagon has ☐ corners.

(b) A shape with 7 sides has ☐ corners.

(c) A circle has ☐ straight sides.

(d) A shape with 8 corners has ☐ sides.

3 **Count each kind of shape. Write how many.**



(a) There are ☐ circles.

(b) There are ☐ triangles.

(c) There are ☐ shapes altogether.

Start here

Keep going. Check your work.

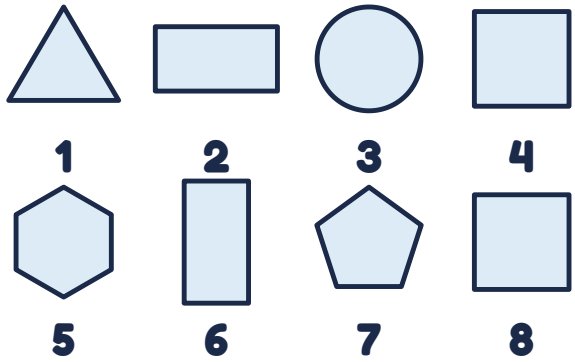
4 **Read each riddle. Circle the shape.**

(a) I have 5 corners. All my sides are straight.
triangle pentagon hexagon

(b) I have 4 corners. All my sides are the same length.
square triangle circle

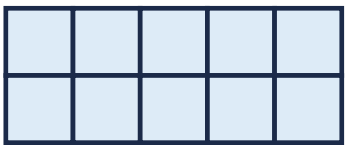
(c) I have 8 straight sides. hexagon octagon pentagon

5 **Look at the shapes. Which shapes have 4 corners?**

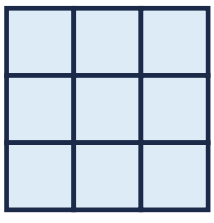


Write their numbers: _____

6 **Look at the 2 grids. Count the small squares.**



Grid A



Grid B

(a) Grid A has ☐ small squares.

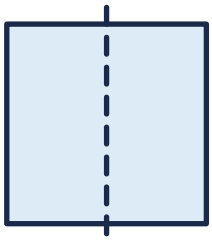
(b) Grid B has ☐ small squares.

(c) Which grid has more small squares? A B

Start here

Nearly there. You can do it.

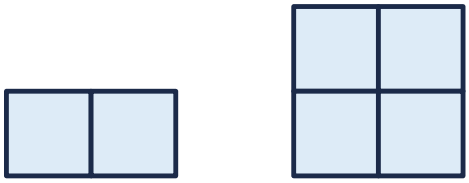
7 **A square is cut along the dashed line.**



(a) Each piece is a triangle square rectangle

(b) Each piece has ☐ corners.

8 **Squares join to make Shape A and Shape B.**

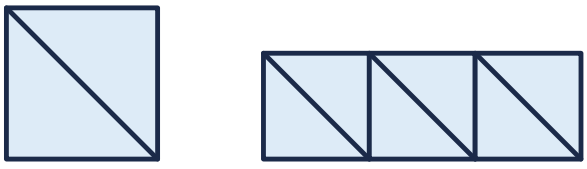


A B

(a) Shape A is a square rectangle triangle

(b) Shape B is a square triangle circle

9 **Count the triangles. Count the small ones.**



A B

(a) Picture A has ☐ triangles.

(b) Picture B has ☐ triangles.

Think harder

Hints are upside down. Try first!

1 **This strip has 4 squares in a row. Count the rectangles in it. A square counts as a rectangle.**



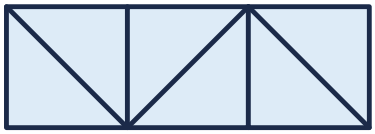
(a) Rectangles that are 1 square long: ☐

(b) Rectangles that are 2 squares long: ☐

(c) All the rectangles in the strip: ☐

Hint. Slide a 2 square rectangle along. Then try 3 long and 4 long.

2 **A strip of 3 squares has 3 lines drawn in it. Count the triangles. Small ones first.**



(a) Small triangles: ☐

(b) All the triangles: ☐

Hint. Do 2 lines that meet make a larger triangle?

3 **Add the sides or the corners.**

(a) 2 shapes have 11 straight sides altogether. One is a pentagon. The other is a square hexagon octagon

(b) A triangle, a square and 1 more shape have 15 corners altogether. The last shape has ☐ corners.

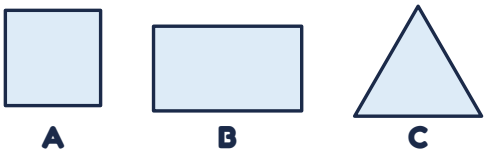
Hint. Add the sides you know. How many are left?

★ Go further

Big challenges. Hints are upside down.

★ Beyond Level 1: Fold lines

1 **Fold a shape so that both halves match exactly. Each way to fold is a fold line. Draw them.**



(a) Square A has ☐ fold lines.

(b) Rectangle B has ☐ fold lines.

(c) Triangle C has 3 sides of the same length. It has ☐ fold lines.

Hint. Imagine a mirror on the line. Both sides must match.

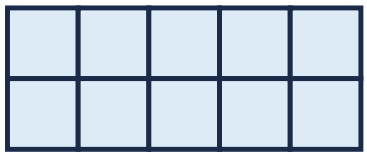
2 **Jenny uses all her square tiles to make a rectangle. A turned rectangle is the same one. A square counts as a rectangle.**

(a) With 12 tiles she can make ☐ different rectangles.

(b) With 9 tiles she can make ☐ different rectangles.

Hint. Try rows of 1, then 2, then 3. Does each row fit exactly?

3 **Count the squares in this grid. Squares of every size count.**



(a) Squares made of 4 small squares: ☐

(b) All the squares: ☐

Hint. Slide a 2 by 2 square along. How many places can it sit?

✔ Check yourself

No hints. Try each one on your own.

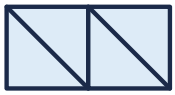
1 A shape has 5 straight sides. It is a **square** **pentagon** **hexagon**

Working: _____

2 How many corners do 3 squares have altogether? ☐

Working: _____

3 How many triangles are in the picture? ☐



4 A shape has 3 more sides than a triangle. It has ☐ sides.

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a grown-up. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

Learn it: Logic and sorting

My goal. I can use clues to find answers. I can sort things and fill in a logic grid.

Clues help us find an answer without guessing. In a logic grid, a tick means yes and a cross means no.

Worked example

Ravi, Jenny and Wei each pick a different drink: milk, juice or tea. Ravi picks tea. Jenny does not pick milk. Who picks what?

	milk	juice	tea
Ravi	✗	✗	✓
Jenny	✗	✓	✗
Wei	✓	✗	✗

1. Ravi picks tea. Tick it. Cross out the rest of that row and column.
2. Jenny does not pick milk. Cross out that box.
3. Jenny's row has 1 empty box left: juice. Wei gets milk.

Answer: Ravi: tea. Jenny: juice. Wei: milk.

How to use a logic grid

1. Read one clue at a time. Tick or cross the boxes.
2. If a row has 1 empty box, tick it.
3. Check every clue at the end.

Your turn

Answer at the back of the book.

Try **Mei, Ben and Kumar each have 1 toy. The toys are a ball, a kite and a drum. Ben has the drum. Mei does not have the ball.**

	ball	kite	drum
Mei			
Ben			
Kumar			

Who has the ball? Mei Ben Kumar

Start here

Do these first. Take your time.

1 **Siti is older than Wei. Wei is older than Jamie.**

(a) The oldest is Siti Wei Jamie

(b) The youngest is Siti Wei Jamie

2 **Ben is 3rd in a queue. Kumar stands just behind Ben. Aisha stands just in front of Ben.**

(a) Kumar is in place ☐.

(b) Aisha is in place ☐.

3 **Ali, Mei, Ben and Siti stand in a line. Ali is 1st. Mei is just behind Ali. Ben is last.**

(a) Siti is in place ☐.

(b) The line from 1st to last: _____

4 **Ali is taller than Mei. Mei is taller than Ravi. Circle True or False.**

(a) Ravi is the shortest. True False

(b) Ali is taller than Ravi. True False

(c) Mei is the tallest. True False

5 **An even number has a last digit of 0, 2, 4, 6 or 8. An odd number has a last digit of 1, 3, 5, 7 or 9. Write each of these numbers in the right list: 4, 7, 10, 13, 16, 9.**

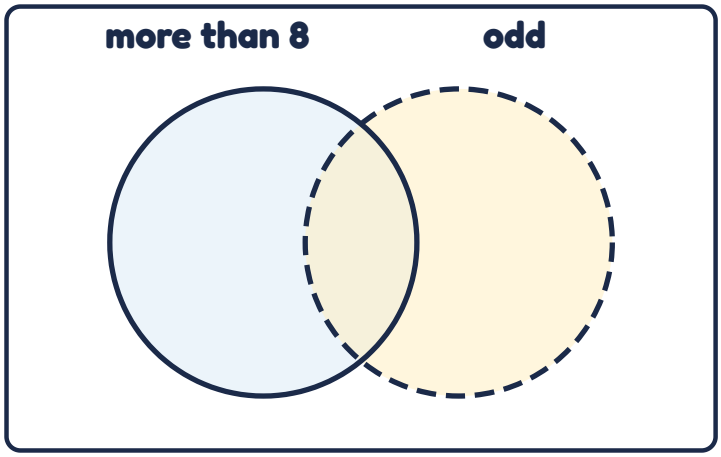
(a) Odd numbers: _____

(b) Even numbers: _____

Start here

Keep going. Check your work.

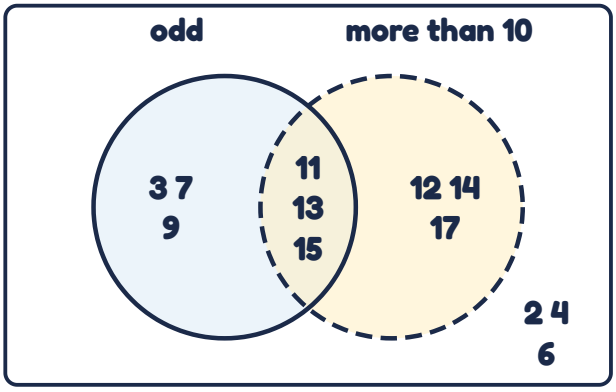
6 Sort the numbers 5 to 14 into the loops. Write each number where it belongs.



(a) How many numbers are in both loops? ☐

(b) Which numbers are outside both loops? _____

7 One number is in the wrong place. Which number is it? ☐



8 Count in 3s and count in 4s from 1 to 24.

(a) Numbers said in both lists: _____

(b) Numbers said in 4s but not in 3s: ☐

Start here

Nearly there. You can do it.

9 **Ravi, Jenny and Wei each pick a different snack. Wei does not pick the bun or the pear. Jenny does not pick the apple. Ravi does not pick the pear.**

	apple	bun	pear
Ravi			
Jenny			
Wei			

(a) Ravi picks the apple bun pear

(b) Jenny picks the apple bun pear

(c) Wei picks the apple bun pear

10 **Circle the odd one out.**

(a) 4 of these numbers are said counting in 4s. Which is not?

8 12 14 16 20

(b) 3 of these shapes have corners. Which does not?

triangle circle square

11 **Find each number.**

(a) A number is between 10 and 20. It is odd. We say it counting in 5s.

What is the number? ☐

(b) A number is even and between 20 and 30. We say it counting in 3s and in 4s. What is the number? ☐

Think harder

Hints are upside down. Try first!

1 **4 children each play a different instrument. Lily does not play the drum, flute or piano. Jason and Aisha do not play the piano. Jason does not play the bell or the drum.**

	drum	flute	piano	bell
Lily				
Jason				
Kumar				
Aisha				

(a) Kumar plays the drum flute piano bell

(b) Aisha plays the drum flute piano bell

Hint. Look for a row with only 1 empty box left. Tick it. Then cross the rest of its column.

2 **Imagine 2 loops. Loop A holds the numbers said counting in 2s. Loop B holds the numbers said counting in 5s. Use the numbers 1 to 20.**

(a) Numbers in both loops: ☐

(b) Numbers in no loop: ☐

Hint. Count in 2s to 20. Then count in 5s to 20. Which numbers are in both lists?

3 **5 friends have 5, 6, 7, 8 and 9 sweets, all different. Mei has the most and Aisha has the fewest. Ravi has fewer sweets than Ben. Ben has fewer than Siti.**

(a) Ben has ☐ sweets.

(b) Siti has ☐ sweets.

Hint. Put the friends in a line from fewest to most. Then match the numbers.

★ Go further

Big challenges. Hints are upside down.

★ Beyond Level 1: Truth and lies

1 **One of 3 children ate the last biscuit. Exactly 1 child is not telling the truth. Who ate the biscuit?** Ali Ben Mei

Ali: "Ben ate it."
Ben: "I did not eat it."
Mei: "Ali ate it."

Hint. Test each child as the one who ate it. Count how many are not telling the truth.

2 **Fill in the grid. Each row and each column has the numbers 1, 2, 3 and 4 once.**

2	1		
4			
3		4	

Hint. Start with a row or column that already has 3 numbers. Which number is missing?

★ Beyond Level 1: Yes or no questions

3 **Mei thinks of a secret number. Ali asks questions. Mei can only answer yes or no.**

(a) The number is from 1 to 4. After "Is it more than 2?" ☐ are left.

(b) The number is from 1 to 8. Ali always finds it with ☐ questions.

(c) The number is from 1 to 16. Ali always finds it with ☐ questions.

Hint. A good question cuts the numbers in half. Keep halving until 1 number is left.

✔ Check yourself

No hints. Try each one on your own.

1 Siti runs faster than Ravi. Ravi runs faster than Jamie. Who is the slowest? Siti Ravi Jamie

Working: _____

2 Use the numbers 1 to 10. How many are even and more than 5? ☐

Working: _____

3 A number is between 20 and 30. It is odd. We say it counting in 5s. What is the number? ☐

Working: _____

4 Ali is taller than Mei. Mei is taller than Ravi. Is Ravi taller than Ali?
Yes No

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a grown-up. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

 Name _____

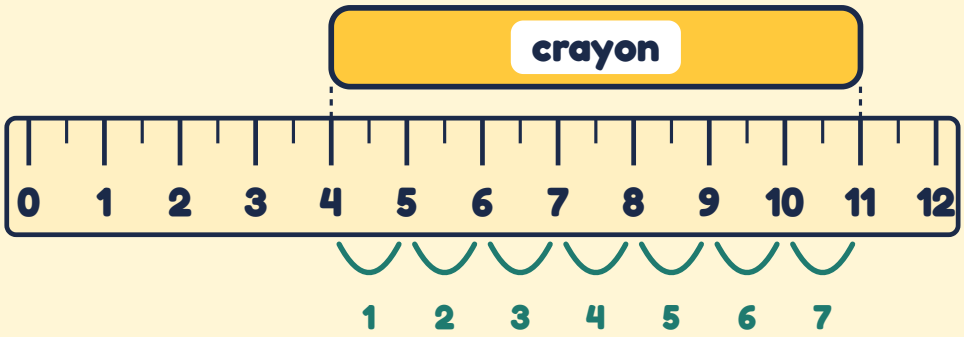
Learn it: Length

My goal. I can compare and measure lengths. I can use a ruler, also when the object does not start at 0.

To measure, use equal units with no gaps. A cube, a clip and a centimetre (cm) are units. On a ruler, count the spaces, not the marks.

Worked example

A crayon lies on a ruler. It starts at the 4 cm mark and ends at the 11 cm mark. How long is the crayon?



1. Find where the crayon starts: the 4 mark. It does not start at 0.
2. Find where it ends: the 11 mark.
3. Count the spaces from 4 to 11. Hop once for each space: 7 hops.
4. Check by subtracting: 11 - 4 = 7.

Answer: The crayon is 7 cm long.

How to measure with a ruler

1. Find the mark where the object starts.
2. Find the mark where it ends.
3. Count the spaces, or subtract: end mark - start mark.

Your turn

Answer at the back of the book.

Try

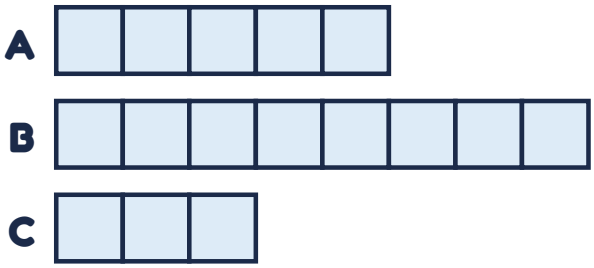
A ribbon is 11 cubes long. A string is 7 cubes long. The ribbon is ☐ cubes longer than the string.

Name _____

Start here

Do these first. Take your time.

1 **Look at the 3 strips of cubes.**



(a) The longest strip is A B C

(b) The shortest strip is A B C

(c) Strip B is ☐ cubes longer than strip A.

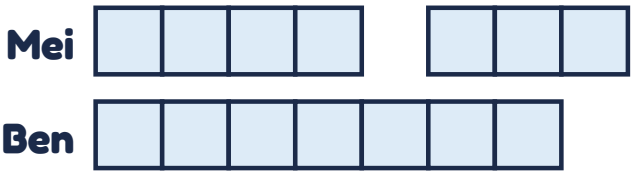
2 **A tape is 13 cubes long. A strip is 8 cubes long. Fill in the blank. The tape is ☐ cubes longer.**

3 **Arrange the lengths in order. Begin with the shortest.**
14 cubes, 9 cubes, 12 cubes, 6 cubes

_____

4 **A rope is 15 cubes long. Ali cuts off 6 cubes. How long is the piece that is left? ☐ cubes**

5 **Mei and Ben measure a pencil with cubes.**



(a) Who measured fairly? Mei Ben

(b) What is wrong in the other row? a gap cubes overlap

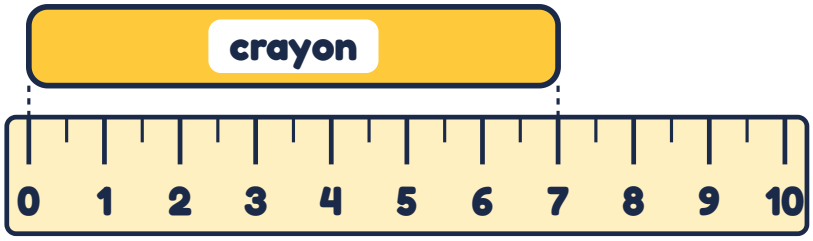
Name _____

Start here

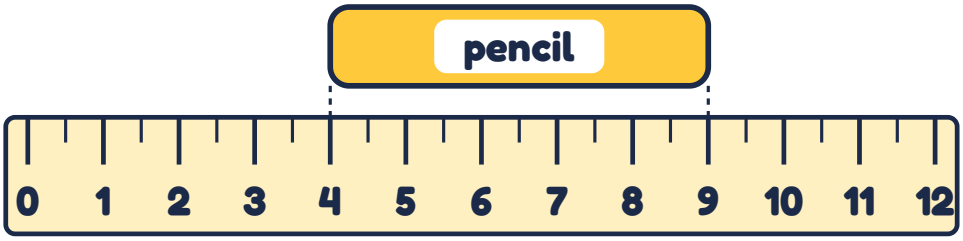
Keep going. Check your work.

6 A tape is 8 cubes long. A strip is 5 cubes longer than the tape. How long is the strip? ☐ cubes

7 Look at the ruler. How long is the crayon? ☐ cm



8 Look at the ruler. How long is the pencil? ☐ cm



9 Look at the ruler. How long is the ribbon? ☐ cm



10 A leaf lies between the 9 cm and 15 cm marks of a ruler. Write its length.

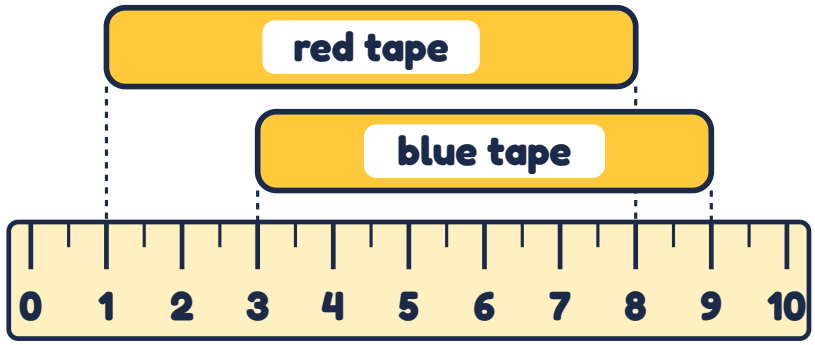
☐ cm

Name _____

Start here

Nearly there. You can do it.

11 Look at the ruler.



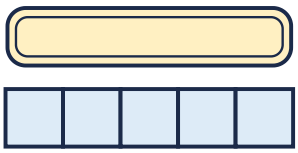
(a) The red tape is ☐ cm long.

(b) The blue tape is ☐ cm long.

(c) The red blue tape is longer.

12 A straw is 6 cm long. Its start is at the 4 cm mark of a ruler. Where is its end? The ☐ cm mark

13 1 stick is as long as 5 cubes. Fill in the blanks.



(a) 3 sticks are as long as ☐ cubes.

(b) 20 cubes are as long as ☐ sticks.

(c) 2 sticks and 3 cubes are as long as ☐ cubes.

14 A table is 6 pencils long. The same table is 9 rubbers long. Which is longer, a pencil or a rubber? a pencil a rubber

Think harder

Hints are upside down. Try first!

1 **A ribbon is 24 cubes long. Ravi cuts it into 4 pieces of equal length.**

(a) Each piece is ☐ cubes long.

(b) Ravi makes ☐ cuts.

Draw the ribbon: _____

Hint. Draw the ribbon as one bar. Mark 4 equal pieces. Count the places you cut.

2 **A clip is the same length as 2 cubes. A pen is 5 clips long. A bag is 2 pens long.**

(a) The pen is ☐ cubes long.

(b) The bag is ☐ cubes long.

Pen: _____

Bag: _____

Hint. First change clips into cubes. Then use the pen to find the bag.

3 **Ali's tape is 10 cubes long. Mei's tape is 4 cubes shorter than Ali's. Ben's tape is 3 cubes longer than Mei's.**

(a) Mei's tape is ☐ cubes long.

(b) Ben's tape is ☐ cubes long.

(c) The longest tape is Ali's Mei's Ben's

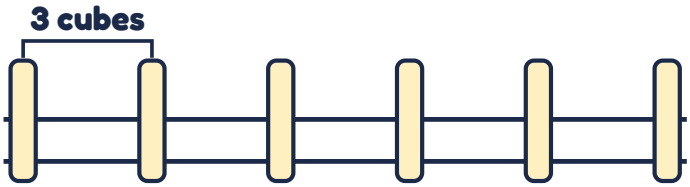
Hint. Find Mei's tape first. Then use Mei's tape to find Ben's.

★ Go further

Big challenges. Hints are upside down.

★ Beyond Level 1: Posts and gaps

1 A fence has 6 posts in a row. Each gap between 2 posts is 3 cubes wide. Ignore how wide a post is.



(a) The fence has ☐ gaps.

(b) From the first post to the last post is ☐ cubes.

(c) A fence with 10 posts has ☐ gaps.

Hint. Count the gaps in the picture, not the posts. How many gaps are there for 2 posts?

2 1 rod is as long as 4 cubes. Ali's rope is 3 rods and 2 cubes long. Mei's rope is 14 cubes long.

(a) Ali's rope is ☐ cubes long.

(b) Which rope is longer? Ali's Mei's They are the same

Ali's rope: _____

Hint. Change the rods into cubes first. Then compare the ropes.

3 A rope is 14 cubes long. Ben cuts it into pieces of 3 cubes or 4 cubes. Nothing is left over. How many pieces of each?

☐ pieces of 4 cubes and ☐ pieces of 3 cubes.

Hint. Try one piece of 4 cubes. Can 3-cube pieces fill what is left? Then try two.

 Name _____

✔ Check yourself

No hints. Try each one on your own.

1 A stick lies from 5 cm to 13 cm. How long is it? ☐ cm

Working: _____

2 Ravi's string is 14 cubes. Lily's is 9. Ravi's is ☐ cubes longer.

Working: _____

3 1 bar is as long as 3 cubes. 5 bars are as long as ☐ cubes.

Working: _____

4 Tape A is 8 cubes. B is 3 cubes longer than A. C is 2 cubes shorter than A.

(a) Which tape is longest? A B C

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a grown-up. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

Learn it: Mixed puzzles

My goal. I can pick a way to start a puzzle: draw it, list it or make a table. I can check and explain my answer.

Puzzles mix what you learned. Do not rush. Read it twice. Then draw it, list it or make a table.

Worked example

Siti's plant is 4 cm tall. It grows 3 cm every week. How tall is the plant after 6 weeks?



1. Say what we know. It starts at 4 cm and grows 3 cm a week.
2. List it. Write 4 first. Then add 3 for each week.
3. The list is 4, 7, 10, 13, 16, 19, 22. Week 6 is the last number.
4. Check: 6 weeks of 3 cm is 3 + 3 + 3 + 3 + 3 + 3 = 18. Then 4 + 18 = 22.

Answer: After 6 weeks the plant is 22 cm tall.

How to start a puzzle

1. Read it twice. What do I know? What must I find?
2. Draw it, list it or make a table.
3. Try. Then check your answer.
4. Say how you know.

Your turn

Answer at the back of the book.

Try **Count in 4s. Fill in the blanks. 4, 8, 12, ☐, ☐**

Name _____

Start here

Do these first. Take your time.

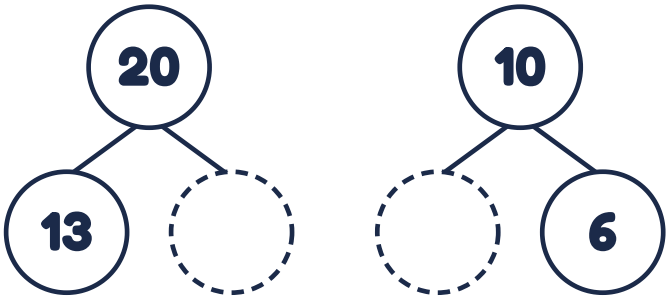
1 **Find the rule. Write the missing number in the tile.**

7	12	17		27

2 **This pattern adds the same number each time. Write the missing numbers in the tiles.**

3	7		15	

3 **Fill in the missing numbers in the bonds.**



4 **Fill in the missing numbers. Both sides must be equal.**

(a) 7 + 6 = 8 + ☐

(b) 9 + 5 = 7 + ☐

Working: _____

Name _____

Start here

Keep going. Check your work.

5 Ravi has 70¢. He buys a drink for 45¢. How much money is left? ☐ ¢

Working: _____

6 Look at the graph. Fill in the blanks.

Cookies sold

Monday	🍪🍪🍪🍪🍪🍪
Tuesday	🍪🍪🍪🍪🍪🍪🍪🍪🍪
Wednesday	🍪🍪🍪🍪
Thursday	🍪🍪🍪🍪🍪🍪🍪

Key: 🍪 = 1 cookie

(a) Tuesday had ☐ more cookies than Wednesday.

(b) Monday and Thursday had ☐ cookies altogether.

7 Count the sides of all the shapes. How many sides are there altogether?

☐ sides



8 Lily is older than Ray. Jamie is younger than Ray. Amelia is older than Lily. Write the names, oldest first.

_____

Draw it: _____

9 Circle the even numbers. 13 16 18 21 24

Start here

Nearly there. You can do it.

10 Make 40¢ with exactly 3 coins. Use 5¢, 10¢, 20¢ or 50¢ coins.

☐ ¢ + ☐ ¢ + ☐ ¢

Try: _____

11 Mei picks a secret number. She adds 7. She gets 15. What number did Mei pick? ☐

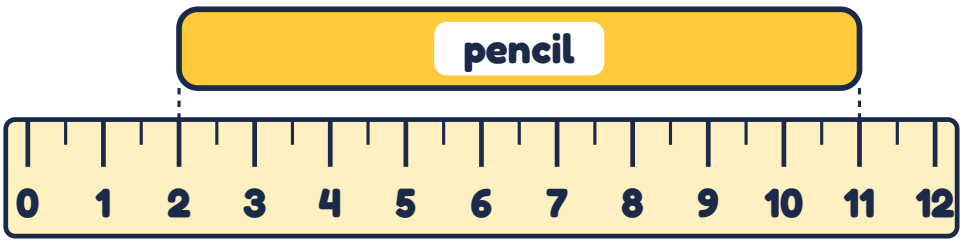
12 Ben picks a secret number. He takes away 3. Then he adds 8. He gets 14. What number did Ben pick? ☐

Undo +8: _____

Undo -3: _____

13 A ribbon is 13 cubes long. A string is 5 cubes shorter than the ribbon. How long is the string? ☐ cubes

14 Look at the ruler. How long is the pencil? ☐ cm



Think harder

Hints are upside down. Try first!

1 **Mei saves 5¢ on day 1, 10¢ on day 2 and 15¢ on day 3. The pattern keeps going.**

(a) On day 4 she saves ☐ ¢.

(b) After 4 days she has saved ☐ ¢ altogether.

Day	Saves	Total so far

Hint. Fill in the table. Add each new amount to the total so far.

2 **Ray, Lily, Jason and Siti sit on chairs 1 to 4. Ray is on chair 4 and Lily is next to Ray. Jason has a lower chair number than Siti. Write the names from chair 1 to chair 4.**

_____

Hint. Put Ray on chair 4 first. Then place Lily. Two names are left.

3 **Ali cuts 4 cubes off a rope, then 3 more cubes. He adds 2 cubes. The rope is now 11 cubes long. How long was it at first?** ☐ **cubes**

Undo +2: _____

Undo -3: _____

Undo -4: _____

Hint. Start at 11 and undo the steps. Begin with the last step.

★ Go further

Big challenges. Hints are upside down.

★ Beyond Level 1: Doubling

1 **Mei folds a paper strip in half. 1 fold makes 2 equal parts. She folds again and again. Each new fold makes twice as many parts.**

(a) 3 folds make ☐ parts.

(b) 5 folds make ☐ parts.

(c) 16 parts need ☐ folds.

Hint. Write the parts after each fold. Double the last number each time.

2 **Aisha, Ben and Mei have 24 stickers altogether. Ben has 4 more than Aisha. Mei has 2 fewer than Ben.**

(a) Aisha has ☐ stickers.

(b) Ben has ☐ stickers.

(c) Mei has ☐ stickers.

Guess: _____

Hint. Guess Aisha's number. Find Ben's and Mei's. Add all 3. Change your guess.

3 **Ravi has 3 triangles and some squares. He counts 17 sides altogether.**

(a) The 3 triangles have ☐ sides.

(b) The squares have ☐ sides.

(c) Ravi has ☐ squares.

Hint. Count the sides of the triangles first. How many sides are left for squares?

Name _____

✔ Check yourself

No hints. Try each one on your own.

1 What are the next 2 numbers? 8, 13, 18, 23, ☐, ☐

Working: _____

2 Fill in the missing number. 20 = 8 + ☐

Working: _____

3 A pen costs 35¢. Ali pays with a 50¢ coin. How much change? ☐ ¢

Working: _____

4 Wei is taller than Siti. Ben is shorter than Siti. Who is the tallest?

Wei Siti Ben

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a grown-up. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Number patterns	☐	☐	☐	☐	/ 4	
2	Number bonds	☐	☐	☐	☐	/ 4	
3	Picture graphs	☐	☐	☐	☐	/ 4	
4	Repeating and growing patterns	☐	☐	☐	☐	/ 4	
5	Working backwards	☐	☐	☐	☐	/ 4	
6	Money	☐	☐	☐	☐	/ 4	
7	2D shapes	☐	☐	☐	☐	/ 4	
8	Logic and sorting	☐	☐	☐	☐	/ 4	
9	Length	☐	☐	☐	☐	/ 4	
10	Puzzle day	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

You finished the Math Inquirers Extension Workbook, Level 1.

Date

Signed by a grown-up

Answers

For grown-ups. Please let your child try each question first, then read the short reason together.

Printed copy: this section is at the end of the book. If you want to keep the answers apart while your child works, take these pages out along the dashed line in the margin.

Unit 1 · Hops and rules

Your turn 8, 13, 18, 23
Why: Add 5 each time: 3 + 5 = 8, 8 + 5 = 13, 13 + 5 = 18, 18 + 5 = 23.

START HERE

1 12, 16
Why: Each hop is 2. 10 and 2 more is 12. 14 and 2 more is 16.

2 25, 35
Why: Each hop is 5. 20 and 5 more is 25. 30 and 5 more is 35.

3 37, 47
Why: Each hop is 10. The tens digit goes up by 1 and the ones digit stays 7.

4 **(a)** 21, 25 **(b)** add 4
Why: From 5 to 9 is 4 more, and from 9 to 13 is 4 more. So the rule is add 4: 17 + 4 = 21, 21 + 4 = 25.

5 **(a)** 9, 6 **(b)** take away 3
Why: The numbers go down by 3 each time: 12 - 3 = 9 and 9 - 3 = 6.

6 add 4
Why: 7 to 11 is 4 more. 11 to 15 is 4 more. 15 to 19 is 4 more.

7 **(a)** Down **(b)** take away 4 **(c)** 14
Why: 30 is more than 26, so the numbers go down. 30 - 26 = 4, so take away 4. 18 - 4 = 14.

8 **(a)** 15 stickers **(b)** 18 stickers
Why: 3, 6, 9, 12, 15, 18. The 4th box has 12, the 5th has 15 and the 6th has 18.

9 **(a)** 7 blocks **(b)** 2
Why: Step 1 has 1 block, Step 2 has 3 and Step 3 has 5. Each step adds 2 blocks, so Step 4 has 5 + 2 = 7.

10 17 is wrong. The right number is 18.
Why: The pattern adds 3 each time: 9, 12, 15, 18, 21, 24. So 17 should be 18.

11 26, 22
Why: Counting back means taking away 2 each time: 28 - 2 = 26 and 24 - 2 = 22.

12 25¢
Why: Each day he puts in 5¢ more: Thursday 20¢, Friday 25¢.

13 3, 7, 11, 15, 19, 23, 27, 31
Why: Start on 3 and count on 4 each time: 3, 7, 11, 15, 19, 23, 27, 31.

14 15, 10, 5, 0
Why: Counting back in 5s means taking away 5 each time: 20 - 5 = 15, 15 - 5 = 10, 10 - 5 = 5, 5 - 5 = 0.

THINK HARDER

1 **(a)** Yes **(b)** No
Why: 2, 7, 12, 17, 22, 27, 32, 37, 42. She lands on 37. She jumps from 37 over 40 to 42.

2 **(a)** 20 and 40 **(b)** 20
Why: Ali says 4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48. Mei says 10, 20, 30, 40, 50. They both say 20 and 40, and 40 - 20 = 20.

3 **(a)** 3 hops **(b)** add 5 **(c)** 5
Why: From the 2nd to the 5th number is 3 hops (2nd to 3rd, 3rd to 4th, 4th to 5th). They make 25 - 10 = 15, so each hop is 15 / 3 = 5. Go back one hop from 10: the 1st number is 5.

GO FURTHER

1 **(a)** 10 dots **(b)** 15 dots **(c)** 28 dots
Why: Each triangle has one more row than the one before, and the new row has one more dot. So the extra dots are 2, 3, 4, 5, 6, 7: 1, 3, 6, 10, 15, 21, 28.

2 7, 22 and 37
Why: Mei says 1, 4, 7, 10, 13, 16, 19, 22, 25, 28, 31, 34, 37, 40. Ali says 2, 7, 12, 17, 22, 27, 32, 37. The same numbers are 7, 22 and 37, and each is 15 more than the one before.

3 1, 2, 3, 6, 9, 18
Why: You need equal hops that make 18 from 2 to 20. Hops of 1 (18 of them), 2 (9), 3 (6), 6 (3), 9 (2) and 18 (1) all work.

CHECK YOURSELF

1 30, 34
Why: The rule is add 4: 26 + 4 = 30 and 30 + 4 = 34.

2 20, 15; take away 5
Why: The numbers go down by 5 each time: 25 - 5 = 20 and 20 - 5 = 15.

3 Yes
Why: Add 4 each time: 6, 10, 14, 18, 22, 26, 30. It lands on 30.

4 27 is wrong. The right number is 26.
Why: The pattern adds 5: 11, 16, 21, 26, 31, 36.

Unit 2 · Make 10 and 20

Your turn 3
Why: 7 and 3 make 10, so 17 and 3 make 20. Check: 17 + 3 = 20.

START HERE

1 **(a)** 6 counters **(b)** 10 = 4 + 6
Why: The frame has 10 spaces. 4 are full, so 6 are empty. 4 + 6 = 10.

2 **(a)** 2 **(b)** 7 **(c)** 5 **(d)** 9
Why: Think of the partner that makes 10: 8 and 2, 7 and 3, 5 and 5, 1 and 9.

3 4 marbles
Why: 6 and 4 make 10, so 4 marbles are in the bag.

4 9, 7, 4, 2
Why: 1 + 9 = 10, 3 + 7 = 10, 6 + 4 = 10, 8 + 2 = 10.

5 5 and 14
Why: First bond: 12 = 7 + 5. Second bond: the two parts 6 and 8 make 14.

6 **(a)** 5 **(b)** 6 **(c)** 2 **(d)** 9
Why: 5 + 5 = 10, 4 + 6 = 10, 8 + 2 = 10, 1 + 9 = 10. So the partners to 20 are 5, 6, 2 and 9.

7 3, 3, 3
Why: The ones digit is 7 each time. 7 + 3 = 10, so each number needs 3 more to reach the next ten.

8 4 books
Why: 16 + 4 = 20, because 6 + 4 = 10.

9 **(a)** 4 **(b)** 5 **(c)** 8
Why: The sign = means both sides have the same value. 6 + 5 = 11, and 7 + 4 = 11. 8 + 2 = 10, and 5 + 5 = 10. 9 + 4 = 13, and 8 + 5 = 13.

10 **(a)** 7, 6, 5 **(b)** down
Why: 2 + 7, 3 + 6 and 4 + 5 all make 9. When one number goes up by 1, the other goes down by 1, so the total stays 9.

11 **(a)** True **(b)** False **(c)** True **(d)** False
Why: 8 + 3 = 11, not 10. 15 + 6 = 21, not 20. The other two are correct bonds.

12 13 boys
Why: 7 + 3 = 10, so 7 + 13 = 20. There are 13 boys.

13 **(a)** 2 stickers **(b)** 12 stickers
Why: 8 + 2 = 10. From 10 to 20 is 10 more, so 8 + 12 = 20.

THINK HARDER

1 6 ways
Why: The ways are 1 and 13, 2 and 12, 3 and 11, 4 and 10, 5 and 9, 6 and 8. 7 and 7 is not allowed because the first bag must have fewer.

2 **(a)** ★ = 6 **(b)** ● = 7
Why: (a) 17 - 5 = 12, and 6 + 6 = 12, so ★ = 6. (b) Three equal numbers make 21: 7 + 7 + 7 = 21, so ● = 7.

3 1 + 2 + 6, 1 + 3 + 5 and 2 + 3 + 4
Why: With 1 as the smallest: 1 + 2 + 6 and 1 + 3 + 5. With 2 as the smallest: 2 + 3 + 4. The smallest number cannot be 3 or more, because 3 + 4 + 5 = 12.

GO FURTHER

1 **(a)** 4 stickers **(b)** 12 stickers
Why: 16 + 8 = 24 stickers. Shared equally, each child has 12. Ali goes from 16 to 12, so he gives 4.

2 **(a)** 11 sweets **(b)** 6 sweets
Why: 17 - 5 = 12. Shared equally, each jar gets 6. Jar A also has the 5 extra: 6 + 5 = 11. Check: 11 + 6 = 17.

3 13, 23 and 33
Why: The numbers that end in 0 between 15 and 45 are 20, 30 and 40. Take away 7 from each: 13, 23 and 33.

4 **(a)** 3, 4, 5, 6, 7, 8, 9 **(b)** 3, 4, 8 and 9
Why: The pairs are 1+2=3, 1+3=4, 1+4=5, 1+5=6, 2+3=5, 2+4=6, 2+5=7, 3+4=7, 3+5=8, 4+5=9. The totals are 3 to 9. 5, 6 and 7 can each be made two ways. 3, 4, 8 and 9 only one way.

CHECK YOURSELF

1 8
Why: 2 and 8 make 10.

2 14
Why: 6 and 4 make 10, so 6 and 14 make 20.

3 3
Why: 7 + 5 = 12. 9 and 3 make 12.

4 9 flowers
Why: 11 + 9 = 20, because 1 + 9 = 10.

Unit 3 · Read and draw graphs

Your turn 3 shells
Why: Ben has 5 shells. Kumar has 2 shells. 5 - 2 = 3. Check: 2 + 3 = 5.

START HERE

1 **(a)** 5 children **(b)** Marbles **(c)** Skipping and tag
Why: Count the faces in each row. Catch has 5. Marbles has 3, the fewest. Skipping and tag both have 7.

2 **(a)** 3 children **(b)** 26 children
Why: (a) "How many more" means subtract: 7 - 4 = 3. (b) Every child chose 1 game, so add every row: 7 + 7 + 5 + 4 + 3 = 26.

3 **(a)** True **(b)** False
Why: (a) Catch has 5 and hopscotch has 4, so it is true. (b) Marbles has 3 faces, not 4: false.

4 We cannot tell
Why: Only 5 children were asked. Other children in the school may choose other games. So we cannot tell.

5 **(a)** Mei **(b)** 3 stickers **(c)** 4 stickers
Why: Mei has 9 stars, the most. Ben has 3. Mei and Siti: 9 - 5 = 4.

6 **(a)** 9 stickers **(b)** 5 stickers
Why: (a) 6 + 3 = 9. (b) Ben had 3 and gets 2 more: 3 + 2 = 5.

7 13 faces
Why: Draw 7 faces for water, 4 for milk and 2 for tea. Count them all: 7 + 4 + 2 = 13.

8 **(a)** Wednesday **(b)** 5 cookies **(c)** 3 cookies
Why: Wednesday has 8 cookies, the most. Thursday has 5. Friday 7 and Monday 4: 7 - 4 = 3.

9 8 books
Why: "3 more than Ali" means add: 5 + 3 = 8. Draw 8 books in Mei's row.

10 5 cakes
Why: The rest is 12 - 7 = 5. Draw 5 pictures in the Cakes row. Check: 7 + 5 = 12.

THINK HARDER

1 **(a)** 11 children **(b)** 5 children
Why: Chess: 5 faces make 10 and the half face makes 11. Singing: 3 faces make 6. 11 - 6 = 5.

2 **(a)** 7 children **(b)** Zoo
Why: 9 + 6 + 8 = 23. From 23 to 30 is 7, so 7 children chose the farm. Zoo has 9 votes, the most.

3 **(a)** 6 children and 10 children **(b)** Graph Q
Why: Graph P has 6 faces and each is 1 child: 6 children. Graph Q has 2 faces and each is 5 children: 5 + 5 = 10. Graph Q has more children, even with fewer pictures.

GO FURTHER

1 **(a)** 5 shells **(b)** 6 shells
Why: Lily: 7 - 2 = 5. Ravi and Lily have 7 + 5 = 12. The rest is 24 - 12 = 12. Jamie and Ray share it equally: 6 each. Check: 7 + 5 + 6 + 6 = 24.

2 **(a)** 5 kites **(b)** 7 balls **(c)** 6 ropes
Why: The extra toys are 2 + 1 = 3. 18 - 3 = 15. Shared equally among 3 rows: 5 each. Kites 5, balls 5 + 2 = 7, ropes 5 + 1 = 6. Check: 5 + 7 + 6 = 18.

3 **(a)** 12 stickers **(b)** 4 pictures
Why: (a) 6 pictures of 2: 2, 4, 6, 8, 10, 12. (b) Count in 3s: 3, 6, 9, 12. That is 4 pictures.

CHECK YOURSELF

1 8 books
Why: Count the books in Kumar's row: 8.

2 5 books
Why: Kumar read 8 and Aisha read 3. 8 - 3 = 5.

3 11 books
Why: 8 + 3 = 11.

4 Aisha
Why: Aisha read 3 books. Kumar read 8 books. 3 is fewer.

Unit 4 · Patterns that go on

Your turn square
Why: The core has 4 shapes. Place 8 ends a core. Place 9 is a circle and place 10 is a square.

START HERE

1 triangle
Why: The core is triangle, circle. It repeats. The last shape is a circle, so a triangle comes next.

2 heart
Why: The core is star, heart, heart. Place 7 is a star. Places 8 and 9 are hearts and place 10 is a star. The 9th shape is a heart.

3 white
Why: The core is dark, dark, white: 3 beads. Count in 3s: 3, 6, 9, 12. Place 12 is the end of a core, so it is white.

4 N, O
Why: The core is M, N, O. After M comes N and then O.

5 clap
Why: The core is clap, clap, stomp: 3 actions. Count in 3s: 3, 6, 9. Place 9 is a stomp. Place 10 starts a new core, so it is a clap.

6 circle
Why: The core is square, star, star, circle: 4 shapes. Count in 4s: 4, 8, 12, 16. Place 16 ends a core, so it is a circle.

7 **(a)** square **(b)** square **(c)** triangle
Why: The core has 3 shapes. Places 9 and 12 end a core (3, 6, 9, 12), so they are squares. Place 13 starts a new core: a triangle.

8 Places 11, 14 and 17
Why: A circle comes once in every core of 3 shapes. So the circles are 3 places apart: 2, 5, 8, 11, 14, 17.

9 **(a)** True **(b)** False **(c)** True
Why: Places 4, 8, 12 end a core, and places 1, 5, 9, 13 start a core. (a) 4 and 8 both end a core: true. (b) 10 is the 2nd shape of a core: false. (c) 13 starts a core: true.

10 8 circles
Why: The core has 3 shapes and 2 are circles. 12 shapes make 4 cores. Count the circles in 2s: 2, 4, 6, 8.

11 **(a)** 11 beads **(b)** Day 7
Why: Day 1: 3. Then add 2 each day: 3, 5, 7, 9, 11, 13, 15. Day 5 is 11 and 15 is day 7.

12 **(a)** 11 blocks **(b)** 17 blocks **(c)** Tower 8
Why: Each tower has 3 more blocks than the one before: 2, 5, 8, 11, 14, 17, 20, 23. Tower 4 has 11 and Tower 6 has 17. 23 is the 8th number, so it is Tower 8.

13 **(a)** 5 straws **(b)** 7 straws **(c)** 9 straws
Why: 1 triangle uses 3 straws. Each new triangle shares a side, so it needs only 2 more straws: 3, 5, 7, 9.

THINK HARDER

1 **(a)** star **(b)** 12 stars **(c)** Place 8
Why: (a) The core has 3 shapes and 15 ends a core, so it is a star. (b) 18 shapes make 6 cores with 2 stars each: 12. (c) The stars are in places 2, 3, 5, 6, 8. The 5th star is in place 8.

2 **(a)** 11 triangles **(b)** 1 straw
Why: The numbers of straws are 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23. 11 triangles use 23 straws. 12 triangles need 25, which is too many. 24 - 23 = 1 straw is left.

3 **(a)** 20 stickers **(b)** Day 10
Why: Add 3 each day: 5, 8, 11, 14, 17, 20, 23, 26, 29, 32. Day 6 is 20. 32 is the 10th number, so it is day 10.

GO FURTHER

1 **(a)** 16 and 22 **(b)** 29
Why: The jumps are 1, 2, 3, 4 and then 5, 6, 7. So 11 + 5 = 16 and 16 + 6 = 22. Then 22 + 7 = 29. The numbers are 1, 2, 4, 7, 11, 16, 22, 29. The 8th is 29.

2 **(a)** 21 straws **(b)** 15 straws **(c)** 6 straws
Why: Ravi: 3, 6, 9, 12, 15, 18, 21. Mei: 3, 5, 7, 9, 11, 13, 15. The gap is 21 - 15 = 6 straws.

3 **(a)** Place 6 **(b)** Place 12
Why: Stars are in even places. Cs are in places 3, 6, 9, 12, 15, 18, 21, 24. The places in both lists are 6, 12, 18 and 24.

CHECK YOURSELF

1 moon
Why: The core is sun, moon, moon: 3 words. Place 9 ends a core (3, 6, 9), so it is a moon.

2 Q
Why: The core has 4 letters. Place 16 ends a core, so place 17 is P and place 18 is Q.

3 19 blocks
Why: Add 4 each time: 3, 7, 11, 15, 19. Tower 5 has 19 blocks.

4 13 straws
Why: Count on in 2s: 3, 5, 7, 9, 11, 13. 6 triangles use 13 straws.

Unit 5 · Back to the start

Your turn 9 beads
Why: She added 6, so undo it by taking away 6: 15 - 6 = 9. Check: 9 + 6 = 15.

START HERE

1 8 counters
Why: Adding 5 made 13. Undo it by taking away 5: 13 - 5 = 8. Check: 8 + 5 = 13.

2 **(a)** 5 **(b)** 8 **(c)** 14 **(d)** 18
Why: Undo the step. Undo add with take away: 13 - 8 = 5 and 15 - 7 = 8. Undo take away with add: 9 + 5 = 14 and 12 + 6 = 18.

3 13 biscuits
Why: She took away 4 and 9 are left. Undo it by adding 4: 9 + 4 = 13. Check: 13 - 4 = 9.

4 8
Why: The arrow adds 9 to reach 17. Go backwards: 17 - 9 = 8. Check: 8 + 9 = 17.

5 9 pencils
Why: Start at the end: 14. Undo buying 8: 14 - 8 = 6. Undo giving 3 away: 6 + 3 = 9. Check: 9 - 3 = 6 and 6 + 8 = 14.

6 18 in the middle, 11 at the start
Why: The last arrow takes away 3 to reach 15. Undo it: 15 + 3 = 18. The first arrow adds 7 to reach 18. Undo it: 18 - 7 = 11.

7 **(a)** 4 **(b)** 13, 13, 19
Why: IN 6 gives OUT 10, so the machine adds 4. IN 9 gives 9 + 4 = 13. OUT 17 came from IN 17 - 4 = 13. IN 15 gives 15 + 4 = 19.

8 **(a)** 9 **(b)** 13 **(c)** 7
Why: Part (a): 15 - 6 = 9. Part (b): 9 + 4 = 13. Part (c): the number taken away is 18 - 11 = 7.

9 floor 9
Why: Undo going up 6: 11 - 6 = 5. Undo going down 4: 5 + 4 = 9. Check: 9 - 4 = 5 and 5 + 6 = 11.

10 8 children
Why: Kumar is the 6th child, so 6 children are at the front, counting him. 2 more are behind him: 6 + 2 = 8 children.

11 7 children
Why: 5 children are at the front, and Aisha is the 5th. The rest are behind her: 12 - 5 = 7.

12 **(a)** 9 **(b)** 15
Why: (a) Ali has more, so Siti has fewer: 15 - 6 = 9. (b) Ben has fewer, so Jenny has more: 9 + 6 = 15.

13 **(a)** True **(b)** False **(c)** True
Why: (a) 9 + 4 = 13, 13 - 6 = 7: True. (b) 12 - 5 = 7, 7 + 3 = 10, not 9: False. (c) 8 + 7 = 15, 15 - 4 = 11: True.

14 **(a)** No **(b)** 9
Why: To undo add 6, take away 6: 15 - 6 = 9. If his number were 21, adding 6 would give 27, not 15. So 21 is not his number.

THINK HARDER

1 **(a)** 17 **(b)** 9 **(c)** 14
Why: Undo the last step first: 13 + 4 = 17. Undo getting 8: 17 - 8 = 9. Undo giving 5 to Ali: 9 + 5 = 14. Check: 14 - 5 = 9, 9 + 8 = 17, 17 - 4 = 13.

2 **(a)** 5 **(b)** 16 **(c)** 25
Why: Adding 9 and taking away 4 is the same as adding 5. IN 11 gives 11 + 5 = 16. For OUT 30, go backwards: 30 - 5 = 25.

3 **(a)** 6 **(b)** 3 **(c)** 10
Why: In front: 7 - 1 = 6. Behind: 4 - 1 = 3. Add Ravi: 6 + 3 + 1 = 10 children. Adding 7 + 4 = 11 would count Ravi twice.

GO FURTHER

1 **(a)** 12 **(b)** 7 **(c)** 8
Why: (b) Undo add 4: 18 - 4 = 14. 7 + 7 = 14, so her number is 7. (c) Undo take away 5: 11 + 5 = 16. 8 + 8 = 16, so her number is 8.

2 **(a)** 9 **(b)** 9 **(c)** 2 **(d)** No
Why: 12 + 6 = 18, 18 - 9 = 9, and 12 - 9 = 3, 3 + 6 = 9. Both give 9. 5 + 6 = 11, 11 - 9 = 2. But you cannot take 9 from 5, so for 5 the order does matter.

3 **(a)** day 5
Why: Start at the top and go backwards. On the last day the frog starts on step 8 and climbs 4 to reach 12. It starts each day 2 steps higher: day 1 starts on 0, day 2 on 2, day 3 on 4, day 4 on 6, day 5 on 8. So it gets out on day 5.

CHECK YOURSELF

1 9
Why: Undo add 7: 16 - 7 = 9.

2 13
Why: Undo getting 9: 17 - 9 = 8. Undo giving away 5: 8 + 5 = 13.

3 15
Why: Undo take away 6 with add 6: 9 + 6 = 15.

4 15
Why: Undo take away 7 with add 7: 8 + 7 = 15. Check: 15 - 7 = 8.

Unit 6 · Pay the exact amount

Your turn 50¢ + 20¢ + 10¢
Why: Take 50¢ first. 30¢ is left. 20¢ + 10¢ = 30¢. Check: 50¢ + 20¢ + 10¢ = 80¢.

START HERE

1 70¢
Why: 50¢ and 10¢ make 60¢. Another 10¢ makes 70¢.

2 $19
Why: $10 and $5 make $15. $2 and $2 make $4. $15 + $4 = $19.

3 **(a)** 5 **(b)** 4 **(c)** 5 **(d)** 2
Why: (a) 10¢ five times is 50¢. (b) 5¢ four times is 20¢. (c) $1 is worth 100¢, and 20¢ five times is 100¢. (d) 50¢ and 50¢ make 100¢, which is $1.

4 **(a)** 20¢ + 5¢ **(b)** 50¢ + 10¢ **(c)** 20¢ + 10¢
Why: (a) 20¢ + 5¢ = 25¢. (b) 50¢ + 10¢ = 60¢. (c) 20¢ + 10¢ = 30¢. Try the largest coin that fits first.

5 20¢
Why: 20¢ + 20¢ = 40¢. Two 10¢ coins make 20¢. Two 50¢ coins make 100¢.

6 **(a)** 40 **(b)** ruler **(c)** 30
Why: (a) 15¢ + 25¢ = 40¢. (b) 45¢ is more than 30¢, so the ruler costs more. (c) 45¢ - 15¢ = 30¢.

7 20¢
Why: Change is what is left: 50¢ - 30¢ = 20¢. Count on to check: 30¢ and 20¢ more make 50¢.

8 Kite and Puzzle
Why: $4 + $8 = $12. Check the others: $7 + $4 = $11, $7 + $8 = $15 and $11 + $4 = $15, so no other pair makes $12.

9 $5
Why: She needs the difference: $13 - $8 = $5. Count on: $8 and $5 more make $13.

10 **(a)** B **(b)** 5
Why: Purse A: 20¢ + 20¢ + 5¢ + 5¢ = 50¢. Purse B: 50¢ + 5¢ = 55¢. B has 55¢ - 50¢ = 5¢ more.

11 **(a)** 2 coins of 20¢ and 0 of 10¢; 1 and 2; 0 and 4 **(b)** 3
Why: Start with the most 20¢ coins. Two 20¢ coins make 40¢. One 20¢ and two 10¢ coins make 40¢. No 20¢ needs four 10¢ coins. That is 3 ways.

12 **(a)** True **(b)** False **(c)** True **(d)** False
Why: (a) 20 + 20 + 10 = 50: True. (b) three 10¢ coins make 30¢: False. (c) ten 10¢ coins make 100¢, which is $1: True. (d) four 5¢ coins make 20¢: False.

13 **(a)** 2 **(b)** 3
Why: (a) 20¢ + 20¢ = 40¢. (b) 50¢ + 20¢ + 20¢ = 90¢. Start with the largest coin each time.

14 50¢, 10¢ and 5¢
Why: 50¢ + 10¢ + 5¢ = 65¢. Taking a 20¢ coin first leaves 45¢, which these coins cannot make.

THINK HARDER

1 50¢ + 5¢ + 5¢ + 5¢ and 20¢ + 20¢ + 20¢ + 5¢
Why: With 50¢, 15¢ is left, made by 5¢ + 5¢ + 5¢. Without 50¢, try 20¢ + 20¢ + 20¢ = 60¢, then 5¢ makes 65¢. A 10¢ coin makes no more sets.

2 **(a)** 4 **(b)** 3
Why: Try 4 coins of 5¢ and 3 coins of 10¢. 4 x 5¢ = 20¢ and 3 x 10¢ = 30¢. 20¢ + 30¢ = 50¢, and 4 + 3 = 7 coins.

3 **(a)** Kite **(b)** Ball
Why: $6 + $9 = $15 and $7 + $8 = $15. The Drum needs $3 more, and no toy costs $3.

GO FURTHER

1 **(a)** 40 **(b)** 20 **(c)** No
Why: (a) 40¢ + 40¢ = 80¢. (b) 20¢ four times is 80¢. (c) Each child would get 25¢. Coins of 20¢ and 10¢ only make 10¢, 20¢, 30¢ and so on, never 25¢.

2 **(a)** 38 **(b)** 20
Why: He spends $50 - $12 = $38 altogether. The bag costs the rest: $38 - $18 = $20. Check: $20 + $18 = $38 and $50 - $38 = $12.

3 **(a)** 6 **(b)** $10, $10, $10, $5, $2, $1
Why: Three $10 notes make $30. $8 is left: $5 + $2 + $1. That is 3 + 3 = 6 pieces. Four $10 notes are too much, and no 5 pieces make $38.

CHECK YOURSELF

1 60¢
Why: 20 + 20 = 40, 40 + 10 = 50, 50 + 5 = 55, 55 + 5 = 60.

2 50¢ + 10¢ + 10¢
Why: 50¢ + 10¢ + 10¢ = 70¢. Without a 50¢ coin, 3 coins make 60¢ at most.

3 25¢
Why: 90¢ - 65¢ = 25¢. Count on: 65¢ and 25¢ more make 90¢.

4 $1
Why: A $1 coin is worth 100¢. 100¢ is more than 85¢, so $1 is more.

Unit 7 · Shape detective

Your turn 3
Why: Touch each pentagon once. There are 3 of them. The circle, the triangle and the square do not have 5 sides.

START HERE

1 **(a)** pentagon **(b)** hexagon **(c)** octagon **(d)** triangle
Why: Count the sides. A has 5 sides: pentagon. B has 6: hexagon. C has 8: octagon. D has 3: triangle.

2 **(a)** 5 **(b)** 7 **(c)** 0 **(d)** 8
Why: (a) A pentagon has 5 sides and 5 corners. (b) The number of corners is the same as the number of sides: 7. (c) A circle has no straight sides: 0. (d) 8 corners means 8 sides.

3 **(a)** 4 **(b)** 3 **(c)** 12
Why: Count one kind at a time. Circles: 4. Triangles: 3. Squares: 4. The hexagon is 1 more. 4 + 3 + 4 + 1 = 12.

4 **(a)** pentagon **(b)** square **(c)** octagon
Why: (a) 5 corners: pentagon. (b) 4 equal sides: square. (c) 8 straight sides: octagon.

5 2, 4, 6 and 8
Why: Shapes 2, 4, 6 and 8 are squares and rectangles. They have 4 corners. The others have 3, 0, 6 or 5 corners.

6 **(a)** 10 **(b)** 9 **(c)** A
Why: (a) 5 in each of 2 rows: 5 + 5 = 10. (b) 3 in each of 3 rows: 3 + 3 + 3 = 9. (c) 10 is more than 9, so Grid A.

7 **(a)** rectangle **(b)** 4
Why: The cut goes from the middle of the top to the middle of the bottom. Each piece is long and thin with 4 square corners: a rectangle.

8 **(a)** rectangle **(b)** square
Why: (a) 2 squares in a row make a long shape with 4 corners: a rectangle. (b) 2 rows of 2 squares make a shape with 4 equal sides: a square.

9 **(a)** 2 **(b)** 6
Why: (a) The line cuts the square into 2 triangles. (b) Each square is cut into 2 triangles. 3 squares make 3 x 2 = 6 triangles. The lines are side by side, so there are no larger triangles.

THINK HARDER

1 **(a)** 4 **(b)** 3 **(c)** 10
Why: 1 square long: 4. 2 squares long: 3. 3 squares long: 2. 4 squares long: 1. Altogether 4 + 3 + 2 + 1 = 10.

2 **(a)** 6 **(b)** 8
Why: There are 6 small triangles. The first 2 lines meet at the bottom and make a larger triangle with the top of the strip. The last 2 lines meet at the top and make another. 6 + 2 = 8 triangles.

3 **(a)** hexagon **(b)** 8
Why: (a) 11 - 5 = 6 sides: a hexagon. (b) 3 + 4 = 7 corners are known. 15 - 7 = 8 corners.

GO FURTHER

1 **(a)** 4 **(b)** 2 **(c)** 3
Why: (a) A square folds across, up and down, and along both corner-to-corner lines: 4. (b) A rectangle folds across and up and down: 2. A corner-to-corner fold does not match. (c) A triangle with 3 equal sides folds from each corner to the middle of the opposite side: 3.

2 **(a)** 3 **(b)** 2
Why: (a) 12 tiles: 12 by 1, 6 by 2 and 4 by 3. That is 3 rectangles. (b) 9 tiles: 9 by 1 and 3 by 3. That is 2 rectangles, and one is a square.

3 **(a)** 4 **(b)** 14
Why: There are 5 + 5 = 10 small squares. (a) A square made of 4 small squares can sit in 4 places along the grid. (b) 10 + 4 = 14. No larger square fits, because the grid is only 2 high.

CHECK YOURSELF

1 pentagon
Why: A shape with 5 sides is a pentagon.

2 12
Why: A square has 4 corners. 4 + 4 + 4 = 12.

3 4
Why: Each square is cut into 2 triangles. 2 + 2 = 4. The 2 lines are side by side, so there is no larger triangle.

4 6
Why: A triangle has 3 sides. 3 + 3 = 6 sides. It is a hexagon.

Unit 8 · Clues and sorting

Your turn Kumar
Why: Ben has the drum. Mei does not have the ball or the drum, so Mei has the kite. The ball is left for Kumar.

START HERE

1 **(a)** Siti **(b)** Jamie
Why: Siti is older than Wei, and Wei is older than Jamie. The order from oldest is Siti, Wei, Jamie.

2 **(a)** 4 **(b)** 2
Why: Just behind 3rd is 4th, so Kumar is 4th. Just in front of 3rd is 2nd, so Aisha is 2nd.

3 **(a)** 3 **(b)** Ali, Mei, Siti, Ben
Why: Ali is 1st and Mei is just behind, so Mei is 2nd. Ben is last, 4th. Siti has the place that is left: 3rd.

4 **(a)** True **(b)** True **(c)** False
Why: The order from tallest is Ali, Mei, Ravi. (a) True: Ravi is last. (b) True: Ali is first and Ravi is last. (c) False: Ali is the tallest.

5 **(a)** 7, 13 and 9 **(b)** 4, 10 and 16
Why: Look at the last digit. 7, 13 and 9 end in 7, 3 and 9: odd. 4, 10 and 16 end in 4, 0 and 6: even.

6 **(a)** 3 **(b)** 6 and 8
Why: More than 8: 9, 10, 11, 12, 13, 14. Odd: 5, 7, 9, 11, 13. In both: 9, 11 and 13. Outside both: 6 and 8, which are not more than 8 and not odd.

7 17
Why: 17 is odd and more than 10, so it belongs in both loops. It is in the "more than 10" loop only.

8 **(a)** 12 and 24 **(b)** 4
Why: In 3s: 3, 6, 9, 12, 15, 18, 21, 24. In 4s: 4, 8, 12, 16, 20, 24. Both: 12 and 24. The 4s list has 6 numbers, and 2 of them are also in the 3s list, so 6 - 2 = 4 are in 4s only.

9 **(a)** bun **(b)** pear **(c)** apple
Why: Wei does not pick the bun or the pear, so Wei picks the apple. Ravi does not pick the pear or the apple, so Ravi picks the bun. Jenny gets the pear.

10 **(a)** 14 **(b)** circle
Why: (a) Counting in 4s we say 4, 8, 12, 16, 20. We do not say 14. (b) A circle has no corners.

11 **(a)** 15 **(b)** 24
Why: (a) Counting in 5s we say 15. It is odd. (b) The even numbers are 22, 24, 26 and 28. Only 24 is said in 3s and in 4s.

THINK HARDER

1 **(a)** piano **(b)** drum
Why: Lily has the bell. Nobody except Kumar can have the piano (Lily, Jason and Aisha cannot), so Kumar has the piano. Jason does not have the bell or the drum, so Jason has the flute. Aisha has the drum.

2 **(a)** 2 **(b)** 8
Why: Said in 2s: 2, 4, 6 ... 20. Said in 5s: 5, 10, 15, 20. In both: 10 and 20. 5s only: 5 and 15. 2s only: 8 numbers, so outside both there are 20 - 2 - 2 - 8 = 8 numbers.

3 **(a)** 7 **(b)** 8
Why: The line is Aisha, Ravi, Ben, Siti, Mei (fewest to most). So Aisha has 5, Ravi 6, Ben 7, Siti 8 and Mei 9.

GO FURTHER

1 Ali
Why: Test each child. If Ali ate it, only Ali is not telling the truth. If Ben ate it, Ben and Mei are both wrong. If Mei ate it, Ali and Mei are both wrong. Only Ali gives exactly 1 wrong child.

2 Row 1: 2 1 3 4. Row 2: 4 3 1 2. Row 3: 1 4 2 3. Row 4: 3 2 4 1.
Why: Column 1 has 2, 4 and 3, so its empty box is 1. Row 1 needs 3 and 4. Column 3 already has a 4, so row 1 ends with 3 and 4. Always fill a box that has only 1 number possible.

3 **(a)** 2 **(b)** 3 **(c)** 4
Why: Each good question cuts the choices in half. 8 numbers become 4, then 2, then 1: 3 questions. 16 numbers become 8, 4, 2, 1: 4 questions.

CHECK YOURSELF

1 Jamie
Why: The order from fastest is Siti, Ravi, Jamie. Jamie is the slowest.

2 3
Why: The even numbers more than 5 are 6, 8 and 10. That is 3 numbers.

3 25
Why: Counting in 5s we say 25. It is between 20 and 30 and it is odd.

4 No
Why: The order from tallest is Ali, Mei, Ravi. Ravi is the shortest, so he is not taller than Ali.

Unit 9 · How long is it?

Your turn 4
Why: 11 - 7 = 4. Check: 7 + 4 = 11.

START HERE

1 **(a)** B **(b)** C **(c)** 3
Why: Count the cubes: A has 5, B has 8, C has 3. B is longest and C is shortest. 8 - 5 = 3, so B is 3 cubes longer than A.

2 5 cubes
Why: "How many longer" asks for the gap. 13 - 8 = 5. Check: 8 + 5 = 13.

3 6, 9, 12, 14
Why: Find the smallest number first (6), then 9, 12 and 14. The shortest length has the fewest cubes.

4 9 cubes
Why: Cutting off takes away. 15 - 6 = 9. Check: 9 + 6 = 15.

5 **(a)** Ben **(b)** a gap
Why: Ben's cubes touch with no gaps, so his row is fair. Mei's row has a gap. A gap makes the pencil look longer than it is, or hides cubes.

6 13 cubes
Why: "Longer than" means add. 8 + 5 = 13. Check: 13 - 8 = 5.

7 7 cm
Why: The crayon starts at the 0 mark and ends at the 7 mark. It is 7 cm long.

8 5 cm
Why: The pencil starts at 4, not at 0. Count the spaces from 4 to 9: 5 spaces. Or subtract: 9 - 4 = 5.

9 9 cm
Why: Start at the 3 mark and end at the 12 mark. 12 - 3 = 9. Check: count 9 spaces.

10 6 cm
Why: Subtract the start from the end: 15 - 9 = 6. Check: from 9, hop 6 spaces to reach 15.

11 **(a)** 7 **(b)** 6 **(c)** red
Why: Red goes from 1 to 8: 8 - 1 = 7 cm. Blue goes from 3 to 9: 9 - 3 = 6 cm. 7 is more than 6, so the red tape is longer.

12 The 10 cm mark
Why: Start at 4 and add the length: 4 + 6 = 10. The straw ends at the 10 cm mark.

13 **(a)** 15 **(b)** 4 **(c)** 13
Why: (a) 3 sticks are 5 + 5 + 5 = 15 cubes. (b) 20 cubes make 4 groups of 5, so 4 sticks. (c) 2 sticks are 10 cubes, and 3 more cubes make 13.

14 a pencil
Why: The table is the same. The table needs fewer pencils, so each pencil is longer. A pencil is longer than a rubber. More pieces does not mean a longer piece.

THINK HARDER

1 **(a)** 6 **(b)** 3
Why: 4 equal pieces: 6 + 6 + 6 + 6 = 24, so each piece is 6 cubes. Cut between the pieces: 3 cuts make 4 pieces. The last piece needs no cut.

2 **(a)** 10 **(b)** 20
Why: The pen is 5 clips, and each clip is 2 cubes: 2 + 2 + 2 + 2 + 2 = 10 cubes. The bag is 2 pens: 10 + 10 = 20 cubes.

3 **(a)** 6 **(b)** 9 **(c)** Ali's
Why: Mei's tape is 10 - 4 = 6 cubes. Ben's tape is 6 + 3 = 9 cubes. 10 is the greatest, so Ali has the longest tape.

GO FURTHER

1 **(a)** 5 **(b)** 15 **(c)** 9
Why: (a) Posts and gaps alternate. The first post has no gap before it, so 6 posts have 5 gaps. (b) 5 gaps of 3 cubes: 3 + 3 + 3 + 3 + 3 = 15. (c) 10 posts have 10 - 1 = 9 gaps. There is always 1 gap fewer than posts.

2 **(a)** 14 **(b)** They are the same
Why: Ali's rope: 3 rods are 4 + 4 + 4 = 12 cubes, and 2 more cubes make 14. Mei's rope is also 14 cubes. The ropes are the same length.

3 Two 4-cube pieces and two 3-cube pieces
Why: One 4-cube piece leaves 10 cubes, and 3-cube pieces cannot fill 10. Two 4-cube pieces leave 6 cubes, and 3 + 3 = 6. So 4 + 4 + 3 + 3 = 14. Three 4-cube pieces leave 2 cubes, which a 3-cube piece cannot fill.

CHECK YOURSELF

1 8 cm
Why: The stick starts at 5 and ends at 13. 13 - 5 = 8.

2 5 cubes
Why: 14 - 9 = 5. Check: 9 + 5 = 14.

3 15 cubes
Why: 5 bars: 3 + 3 + 3 + 3 + 3 = 15 cubes.

4 **(a)** B
Why: B is 8 + 3 = 11 cubes. C is 8 - 2 = 6 cubes. A is 8 cubes. 11 is the greatest, so B is longest.

Unit 10 · Mix and solve

Your turn 16, 20
Why: Each number is 4 more. 12 + 4 = 16 and 16 + 4 = 20.

START HERE

1 22
Why: Each tile is 5 more than the one before: 7, 12, 17. So the missing tile is 17 + 5 = 22. Check: 22 + 5 = 27.

2 11, 19
Why: From 3 to 7 is 4 more, so the rule is add 4. After 7 comes 11. After 11 comes 15. After 15 comes 19.

3 7 and 4
Why: 13 and 7 make 20, because 3 and 7 make 10. 6 and 4 make 10.

4 **(a)** 5 **(b)** 7
Why: (a) 7 + 6 = 13. 8 and 5 make 13. (b) 9 + 5 = 14. 7 and 7 make 14.

5 25¢
Why: 70¢ - 45¢ = 25¢. Check: 45¢ + 25¢ = 70¢.

6 **(a)** 5 **(b)** 13
Why: Tuesday has 9 cookies and Wednesday has 4. 9 - 4 = 5 more. Monday has 6 and Thursday has 7. 6 + 7 = 13.

7 10 sides
Why: A triangle has 3 sides and a square has 4 sides. 3 + 4 + 3 = 10.

8 Amelia, Lily, Ray, Jamie
Why: Amelia is older than Lily, Lily is older than Ray, and Ray is older than Jamie. The order is Amelia, Lily, Ray, Jamie.

9 16, 18, 24
Why: An even number can be shared into 2 equal groups with none left. 16, 18 and 24 are even. 13 and 21 are odd.

10 20¢ + 10¢ + 10¢
Why: 20¢ + 10¢ + 10¢ = 40¢. This is the only way: 50¢ is too big, and 5¢ coins would need 35¢ from 2 coins, which no 2 coins make.

11 8
Why: Go backwards. Undo adding 7 by taking away 7: 15 - 7 = 8. Check: 8 + 7 = 15.

12 9
Why: Go backwards. Undo the last step: 14 - 8 = 6. Undo the first step: 6 + 3 = 9. Check: 9 - 3 = 6 and 6 + 8 = 14.

13 8 cubes
Why: "Shorter than" means take away. 13 - 5 = 8. Check: 8 + 5 = 13.

14 9 cm
Why: The pencil starts at 2 and ends at 11. 11 - 2 = 9. Check: count 9 spaces.

THINK HARDER

1 **(a)** 20 **(b)** 50
Why: The amount grows by 5¢ each day: 5, 10, 15, 20. On day 4 she saves 20¢. Add each day to the total: 5 + 10 = 15, 15 + 15 = 30, 30 + 20 = 50.

2 Jason, Siti, Lily, Ray
Why: Ray is on chair 4, so Lily is on chair 3. Jason and Siti have chairs 1 and 2. Jason has the lower number, so Jason is on chair 1 and Siti on chair 2.

3 16 cubes
Why: Go backwards. Undo adding 2: 11 - 2 = 9. Undo cutting off 3: 9 + 3 = 12. Undo cutting off 4: 12 + 4 = 16. Check: 16 - 4 = 12, 12 - 3 = 9, 9 + 2 = 11.

GO FURTHER

1 **(a)** 8 **(b)** 32 **(c)** 4
Why: Parts after each fold: 2, 4, 8, 16, 32. (a) 3 folds make 8 parts. (b) 5 folds make 32 parts. (c) 16 parts need 4 folds.

2 **(a)** 6 **(b)** 10 **(c)** 8
Why: Try 6 for Aisha. Ben has 6 + 4 = 10. Mei has 10 - 2 = 8. Together 6 + 10 + 8 = 24. A smaller guess gives less than 24 and a greater guess gives more than 24.

3 **(a)** 9 **(b)** 8 **(c)** 2
Why: A triangle has 3 sides: 3 + 3 + 3 = 9. The squares have 17 - 9 = 8 sides. A square has 4 sides, and 4 + 4 = 8, so there are 2 squares.

CHECK YOURSELF

1 28, 33
Why: The rule is add 5. 23 + 5 = 28 and 28 + 5 = 33.

2 12
Why: 8 and 2 make 10, and 10 and 10 make 20. So 8 and 12 make 20.

3 15¢
Why: 50¢ - 35¢ = 15¢. Check: 35¢ + 15¢ = 50¢.

4 Wei
Why: Wei is taller than Siti, and Siti is taller than Ben. So Wei is the tallest.